Dynamics of Rigid Bodies

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Syllabus

- **Dynamics of Rigid bodies [12 Lectures]:** Translation/Rotation of rigid bodies, Chasles' theorem, time derivative of vector for different references. Parallel axis theorems, Rotational Pure rotation of a body of revolution about its axis of revolution/combined with translation. Three dimensional rotation, moment of inertia tensor, relation between angular momentum and torque in three dimensions, Motion in non-inertial frames, centrifugal force, Coriolis force/acceleration, rate of change of vector in inertial and rotating frames. **[Dr. Amrita Puri]**

Rigid Body Kinematics

- **Particle Kinematics:** Development of the relationships governing the displacement, velocity, and acceleration of points as they moved along straight or curved paths
- **Rigid Body Kinematics:** One must also account for the rotational motion of the body. Thus, rigid-body kinematics involves both linear and angular displacements, velocities, and accelerations

Why study Rigid Body Kinematics?

- **Design based on motion and calculation of forces due to the motion:** we frequently need to generate, transmit, or control certain motions by the use of cams, gears, and linkages of various types. Here we must analyze the displacement, velocity, and acceleration of the motion to determine the design geometry of the mechanical parts. Furthermore, as a result of the motion generated, forces may be developed which must be accounted for in the design of the parts.
- **Calculation of motion based on action of forces:** Second, we must often determine the motion of a rigid body caused by the forces applied to it. Calculation of the motion of a rocket under the influence of its thrust and gravitational attraction is an example of such a problem.

Ideal Rigid Body and Assumption of Rigidity

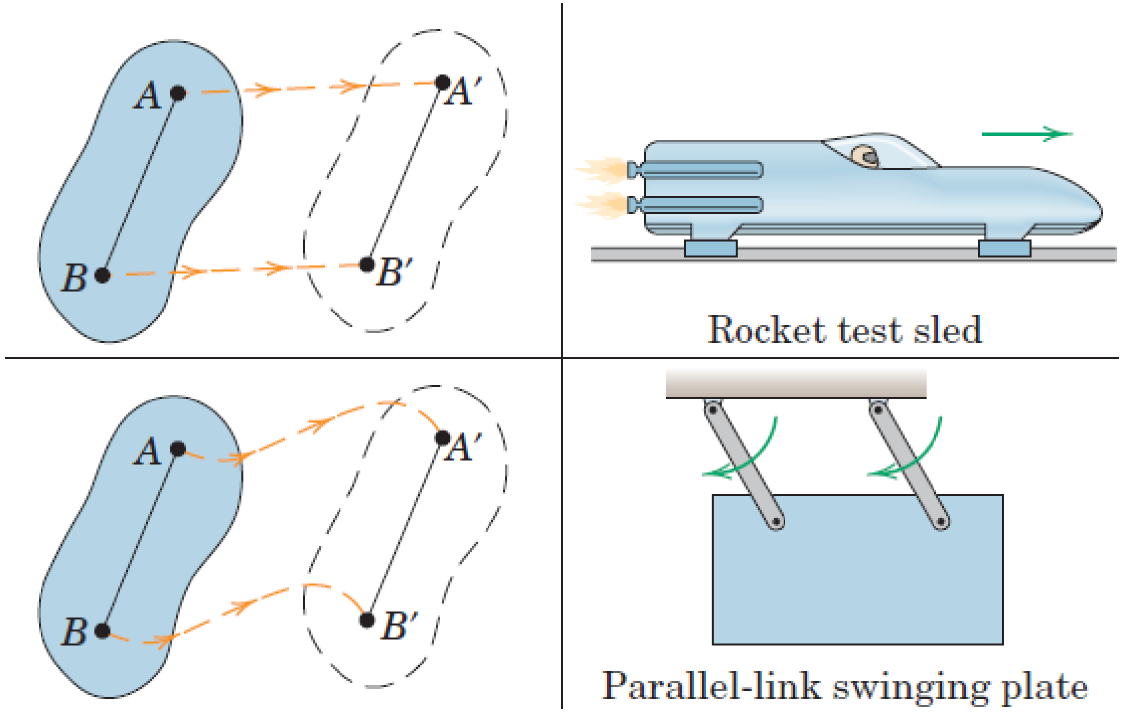
- A *rigid body* is defined as a system of particles for which the distances between the particles remain unchanged
- If each particle of such a body is located by a position vector from reference axes attached to and rotating with the body, there will be no change in any position vector as measured from these axes.
- If the movements associated with the changes in shape are very small compared with the movements of the body as a whole, then the assumption of rigidity is usually acceptable
- The displacements due to the flutter of an aircraft wing, for instance, do not affect the description of the flight path of the aircraft as a whole, and thus the rigid-body assumption is clearly acceptable
- the problem is one of describing, as a function of time, the internal wing stress due to wing flutter, then the relative motions of portions of the wing cannot be neglected, and the wing may not be considered a rigid body.

Plane Motion

- A rigid body executes plane motion when all parts of the body move in parallel planes. For convenience, we generally consider the *plane of motion* to be the plane which contains the mass center, and we treat the body as a thin slab whose motion is confined to the plane of the slab.

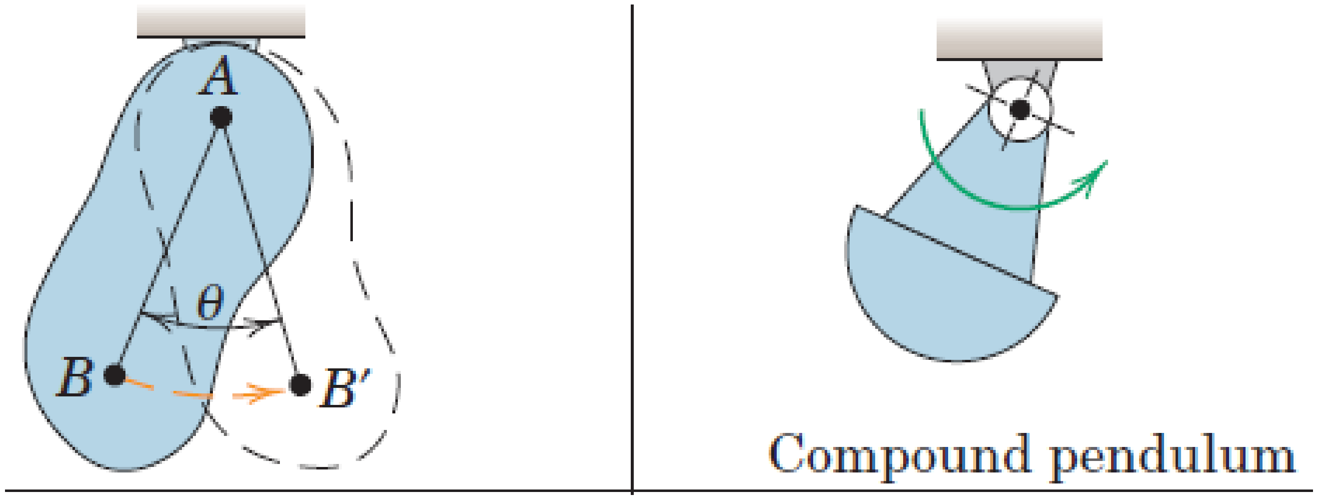
Translation in Plane Motion

- ***Translation*** is defined as any motion in which every line in the body remains parallel to its original position at all times. In translation there is *no rotation of any line in the body*. In *rectilinear translation*, all points in the body move in parallel straight lines.
- In *curvilinear translation*, all points move on congruent curves.
- We note that in each of the two cases of translation, the motion of the body is completely specified by the motion of any point in the body, since all points have the same motion. Thus, our earlier study of the motion of a point (particle) in Chapter 2 enables us to describe completely the translation of a rigid body.



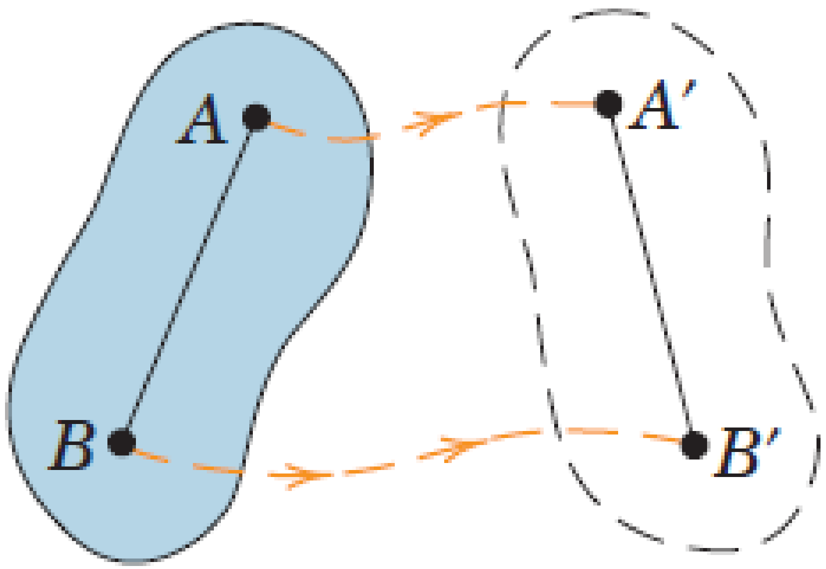
Rotation in Plane Motion

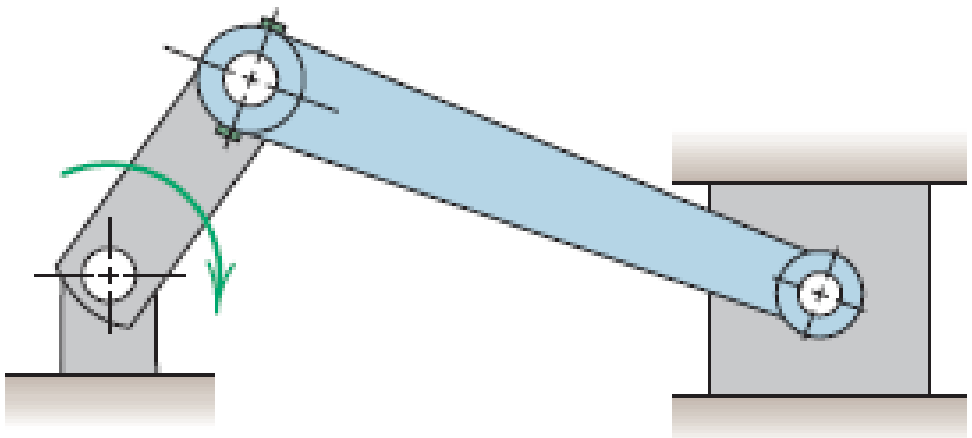
- *Rotation about a fixed axis is the angular motion about the axis. It follows that all particles in a rigid body move in circular paths about the axis of rotation, and all lines in the body which are perpendicular to the axis of rotation (including those which do not pass through the axis) rotate through the same angle in the same time. Again, our discussion in Chapter 2 on the circular motion of a point enables us to describe the motion of a rotating rigid body, which is treated in the next article.*



General Plane Motion

- ***General plane motion*** of a rigid body is a combination of translation and rotation. We will utilize the principles of relative motion covered in Art. 2/8 to describe general plane motion

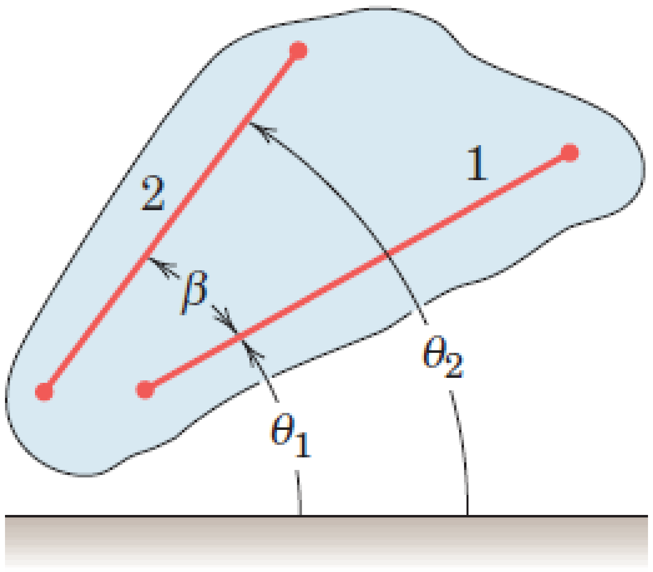




Connecting rod in a reciprocating engine

Rotation

- The rotation of a rigid body is described by its angular motion
- All lines on a rigid body in its plane of motion have the same angular displacement, the same angular velocity, and the same angular acceleration.
- The angular positions of any two lines 1 and 2 attached to the body are specified by θ_1 and θ_2 measured from any convenient fixed reference direction. Because the angle β is invariant, the relation $\beta = \theta_2 - \theta_1$ upon differentiation with respect to time gives $\dot{\theta}_2 = \dot{\theta}_1$ and $\ddot{\theta}_2 - \ddot{\theta}_1$, during a finite interval, $\Delta\theta_1 = \Delta\theta_2$.
- Angular motion of a line depends only on its angular position with respect to any arbitrary fixed reference and on the time derivatives of the displacement. Angular motion does not require the presence of a fixed axis, normal to the plane of motion, about which the line and the body rotate.



Angular Motion Relations

The angular velocity ω and angular acceleration α of a rigid body in plane rotation are, respectively, the first and second time derivatives of the angular position coordinate θ of any line in the plane of motion of the body.

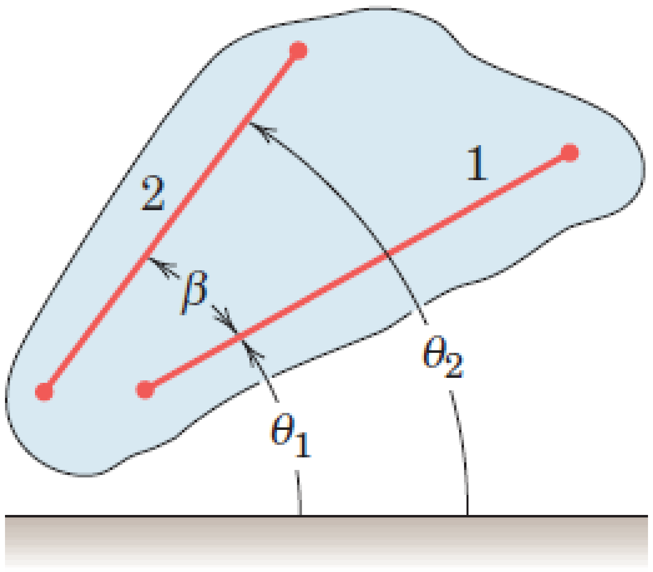
$$\omega = \frac{d\theta}{dt} = \dot{\theta}$$

$$\alpha = \frac{d\omega}{dt} = \dot{\omega} \qquad \alpha = \frac{d^2\theta}{dt^2} = \ddot{\theta}$$

$$\alpha d\theta = \frac{d\omega}{dt} d\theta = \omega d\omega \qquad \alpha d\theta = \frac{d^2\theta}{dt^2} d\theta = \ddot{\theta} d\theta$$

$$\alpha d\theta = \omega d\omega = \ddot{\theta} d\theta$$

$$\dot{\theta} d\dot{\theta} = \ddot{\theta} d\theta$$



Constant Angular Acceleration

For constant angular acceleration,

Integrating $\alpha = \dfrac{d\omega}{dt}$ $\Longrightarrow$ $\omega = \omega_0 + \alpha t$

Integrating $\alpha d\theta = \omega d\omega$ $\Longrightarrow$ $\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$

Using $\omega = \omega_0 + \alpha t$

$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$ $\Longrightarrow$ $\theta = \theta_0 + \omega_0 t + \dfrac{1}{2}\alpha t^2$

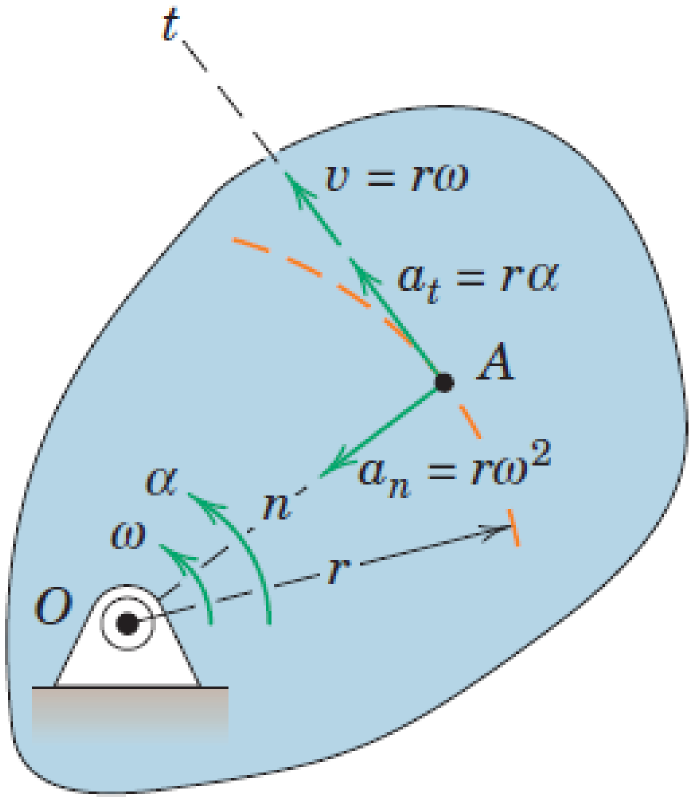
Rotation about a Fixed Axis

- When a rigid body rotates about a fixed axis, all points other than those on the axis move in concentric circles about the fixed axis. Thus, for the rigid body rotating about a fixed axis normal to the plane of the figure through *O*, any point such as *A* moves in a circle of radius *r*.
- Relationships between the linear motion of A and the angular motion of the line normal to its path, which is also the angular motion of the rigid body.

$$v = r\omega$$

$$a_n = r\omega^2 = \frac{v^2}{r} = v\omega$$

$$a_t = r\alpha$$



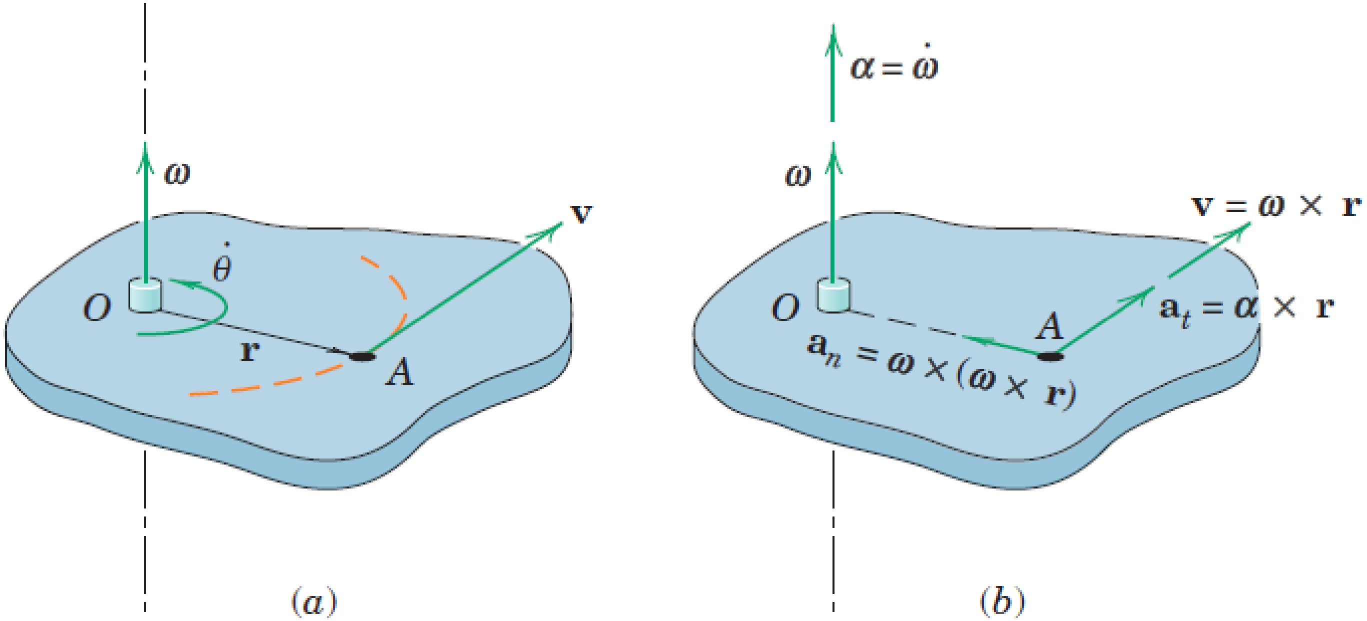
Vector Relations

The angular velocity of the rotating body may be expressed by the vector $\boldsymbol{\omega}$ normal to the plane of rotation and having a sense governed by the right-hand rule.

$$V = \dot{r} = \omega \times r$$

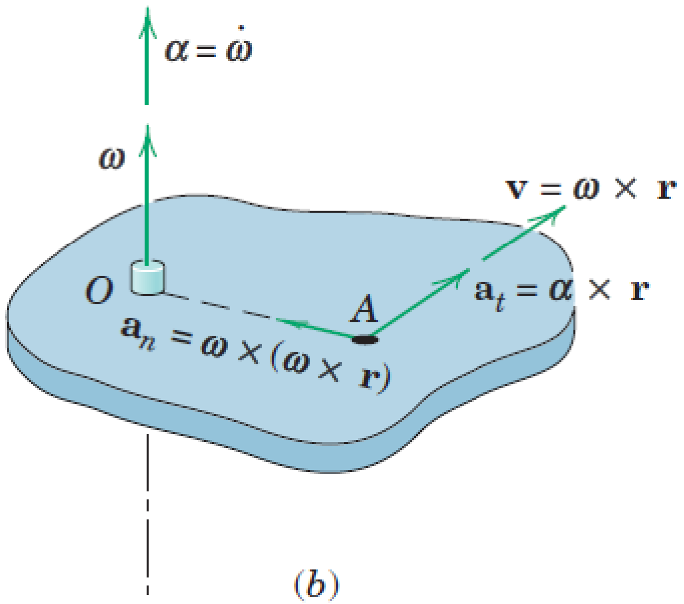
Order of the vectors to be crossed must be retained

$$r \times \omega = -v$$



Vector Relations

Acceleration of point A is obtained by differentiating the cross product expression for v



(b)

$$a = \dot{v} = \omega \times \dot{r} + \dot{\omega} \times r$$

$$a = \omega \times (\omega \times r) + \dot{\omega} \times r$$

$$a = \omega \times v + \alpha \times r$$

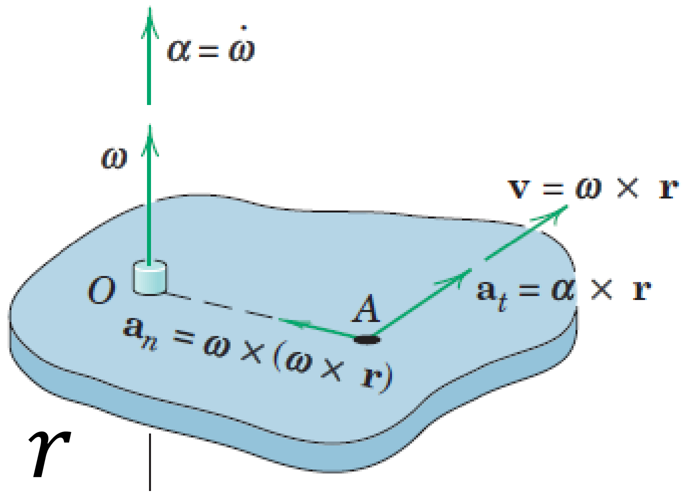
$\alpha = \dot{\omega}$ Angular acceleration of point A

$$v = \omega \times r$$

$$a_n = \omega \times (\omega \times r)$$

$$a_t = \alpha \times r$$

Vector Relations



Acceleration of point A is obtained by differentiating the cross product expression for v

$$a = \dot{v} = \omega \times \dot{r} + \dot{\omega} \times r$$

$$a = \omega \times (\omega \times r) + \dot{\omega} \times r$$

$$a = \omega \times v + \alpha \times r$$

$$v = \omega \times r$$

$$\alpha_n = \omega \times (\omega \times r)$$

$$\alpha_t = \alpha \times r$$

$\alpha = \dot{\omega}$ Angular acceleration of point A

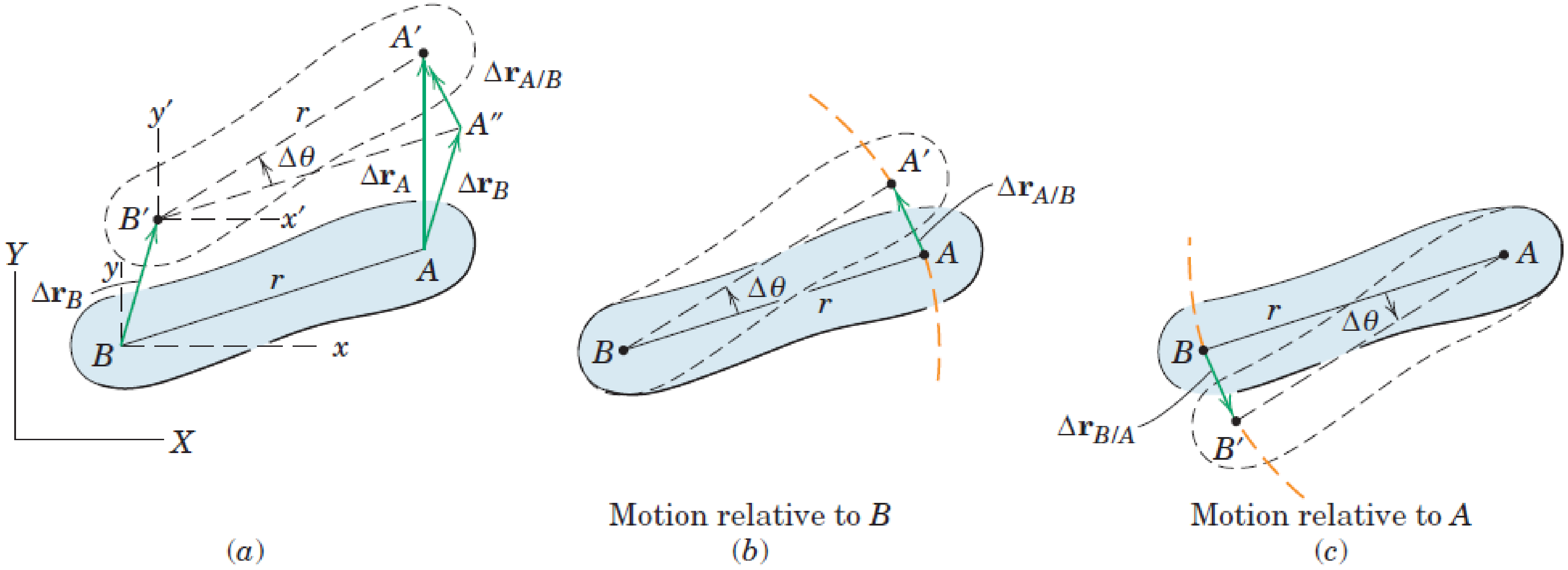
- For three-dimensional motion of a rigid body, the angular-velocity vector may change direction as well as magnitude, and in this case, the angular acceleration, α which is the time derivative of angular velocity, $\dot{\omega}$,will no longer be in the same direction as ω.

Absolute Motion

Acceleration of point A is obtained by differentiating the cross product expression for v

Relative Motion

- Two points on the same rigid body for our two particles
- The consequence of this choice is that the motion of one point as seen by an observer translating with the other point must be circular since the radial distance to the observed point from the reference point does not change.



Relative Motion

$$\Delta r_{B/A} = -\Delta r_{A/B}$$

$$\Delta r_A = \Delta r_B + \Delta r_{A/B}$$

$$V_A = V_B + V_{A/B}$$

$$v_{A/B} = \lim_{\Delta t \to 0} \left(\left| \Delta r_{A/B} \right| / \Delta t \right) = \lim_{\Delta t \to 0} \left(r \Delta \theta / \Delta t \right)$$

$$v_{A/B} = r\omega$$

Relative linear velocity is always perpendicular to the line joining the two points in question

$$v_{A/B} = \omega \times r$$

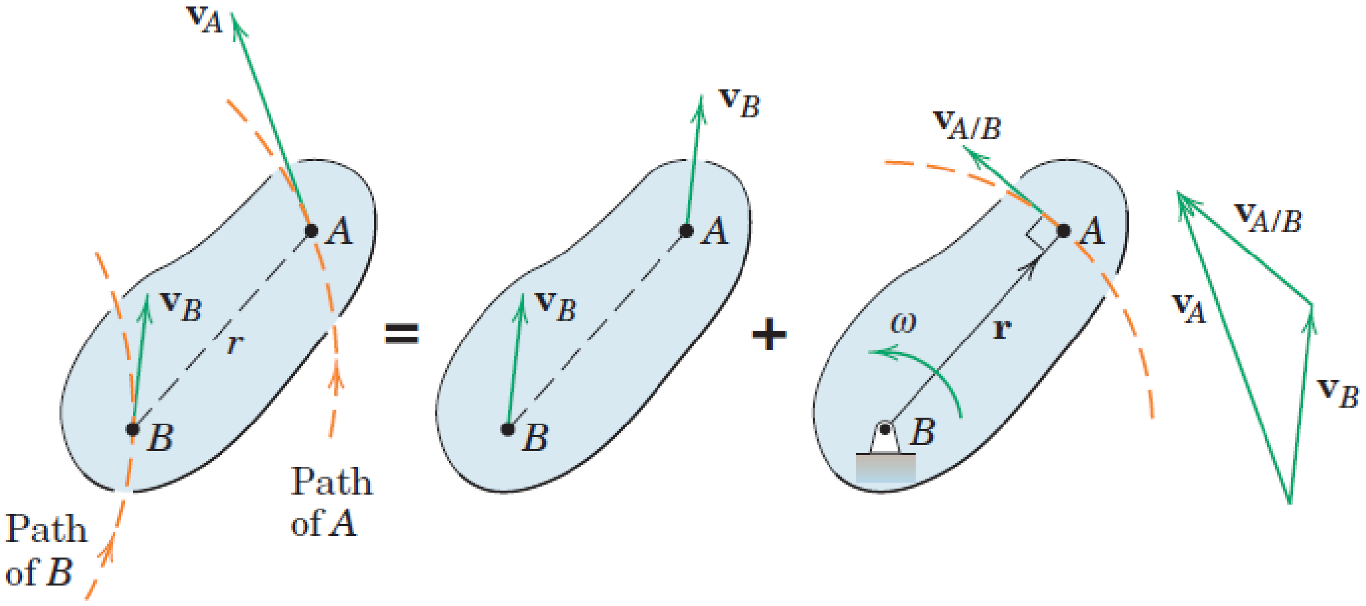
Relative Motion

$$V_A = V_B + V_{A/B}$$

$$v_{A/B} = r\omega$$

$$v_{A/B} = \omega \times r$$

- With B chosen as the reference point, the velocity of A is the vector sum of the translational portion $\boldsymbol{V_B}$, plus the rotational portion $\boldsymbol{V_{A/B}}$, which has the magnitude $v_{A/B} = r\omega$, where ω is the *absolute* angular velocity of *AB*.
- The fact that **the *relative linear velocity* is *always perpendicular*** to the line joining the two points in question is an important key to the solution of many problems



Instantaneous Center of Zero Velocity

Choosing a unique reference point which momentarily has zero velocity

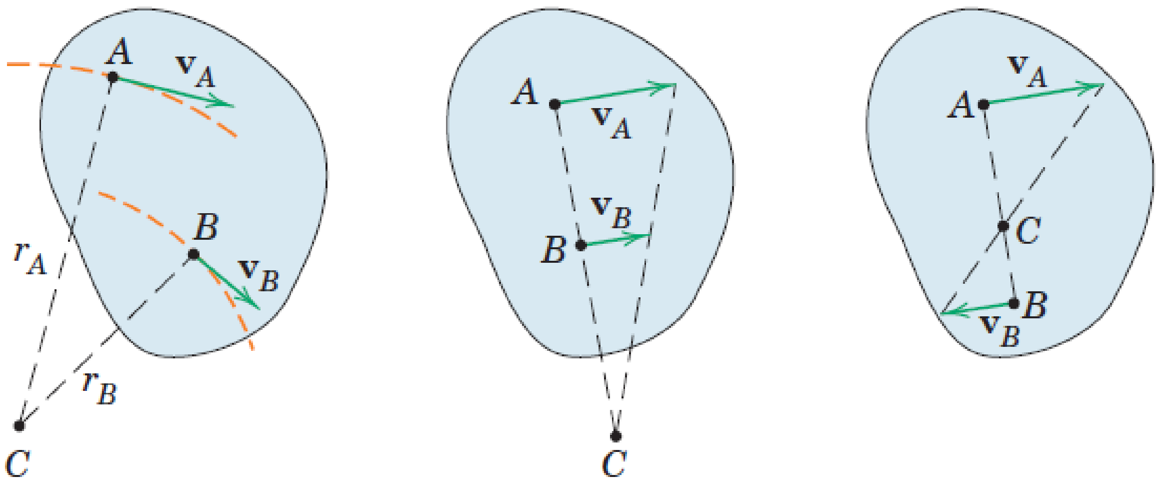
As far as velocities are concerned, the body may be considered to be in pure rotation about an axis, normal to the plane of motion, passing through this point.

This axis is called the instantaneous axis of zero velocity.

The intersection of this axis with the plane of motion is known as the instantaneous center of zero velocity.

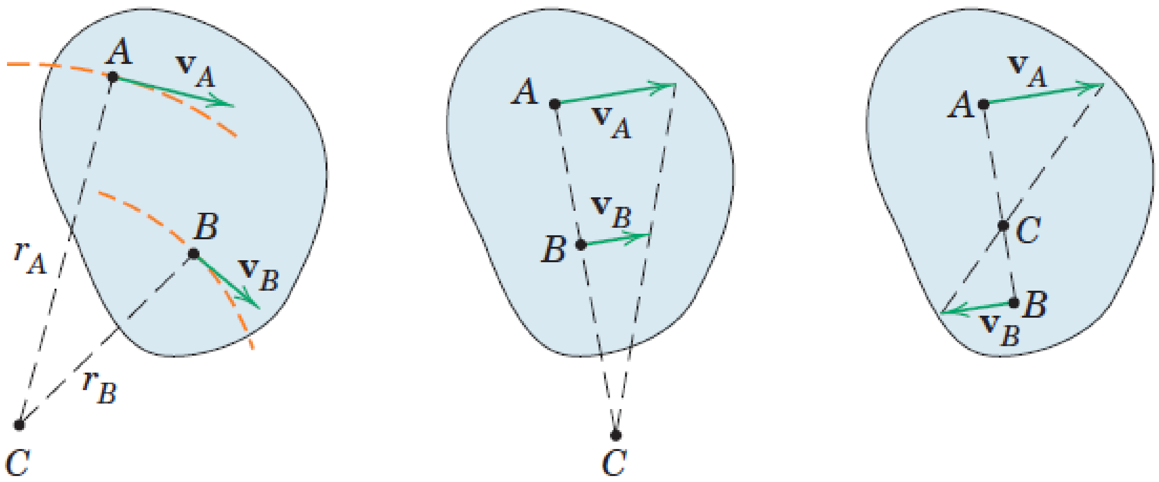
Instantaneous Center of Zero velocity

- Directions of the absolute velocities of any two points A and B on the body are known and are not parallel
- If there is a point about which *A* has absolute circular motion at the instant considered, this point must lie on the normal to v_A through *A*. Similar reasoning applies to *B*, and the intersection of the two perpendiculars fulfills the requirement for an absolute center of rotation *at the instant considered*.
- Point *C* is the instantaneous center of zero velocity and may lie on or off the body.
- If it lies off the body, it may be visualized as lying on an imaginary extension of the body.
- The instantaneous center need not be a fixed point in the body or a fixed point in the plane.



Instantaneous Center of Zero velocity

- If we also know the magnitude of the velocity of one of the points, say, v_A, we may easily obtain the angular velocity ω of the body and the linear velocity of every point in the body.
- Once the instantaneous center is located, the direction of the instantaneous velocity of every point in the body is readily found since it must be perpendicular to the radial line joining the point in question with C.

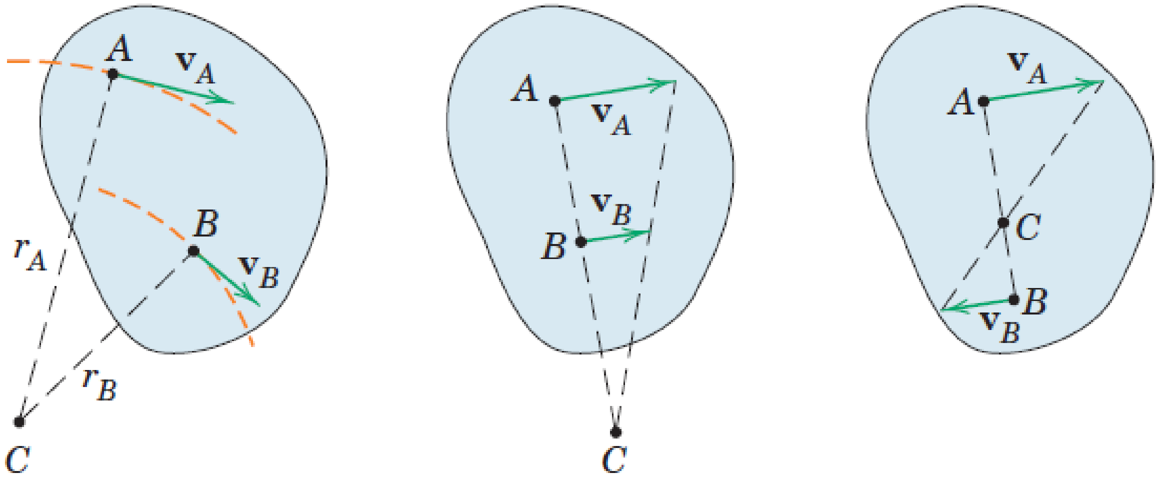


$$\omega = \frac{v_A}{r_A}$$

$$v_B = r_B\omega = (r_B/r_A)v_A$$

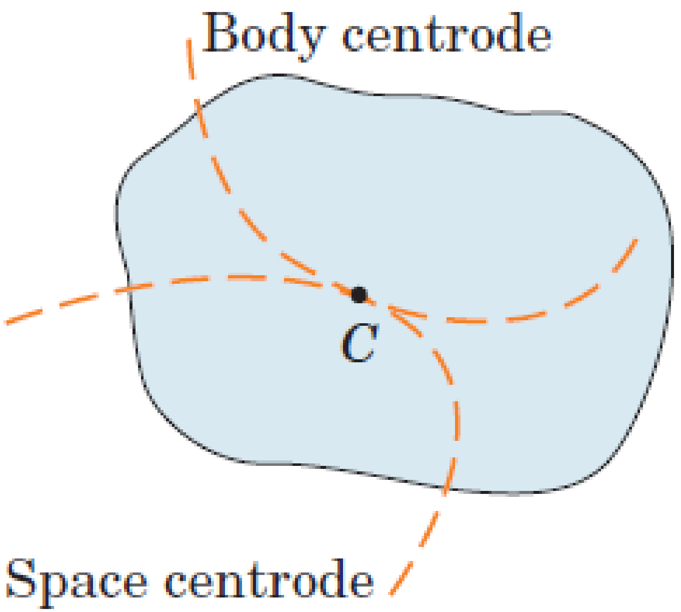
Instantaneous Center of Zero velocity

- If the **velocities of two points in a body having plane motion are parallel** and the line joining the points is perpendicular to the direction of the velocities, the instantaneous center is located by direct proportion.
- As the **parallel velocities become equal in magnitude**, the instantaneous center moves farther away from the body and approaches infinity in the limit as the body stops rotating and translates only.



Motion of Instantaneous Center of Zero velocity

- As the body changes its position, the instantaneous center C also changes its position both in space and on the body.
- The **locus of the instantaneous centers in space** is known as the **space centrode.**
- The **locus of the positions of the instantaneous centers on the body** is known as the **body centrode**.
- At the instant considered, the two curves are tangent at the position of point C.



Motion of Instantaneous Center of Zero velocity

- Although the instantaneous center of zero velocity is momentarily at rest, its acceleration generally is not zero. Thus, this point may not be used as an instantaneous center of zero acceleration in a manner analogous to its use for finding velocity.

- An instantaneous center of zero acceleration does exist for bodies in general plane motion.

Relative Acceleration

- The acceleration of point A equals the vector sum of the acceleration of point B and the acceleration which A appears to have to a nonrotating observer moving with B

$$v_A = v_B + v_{A/B}$$

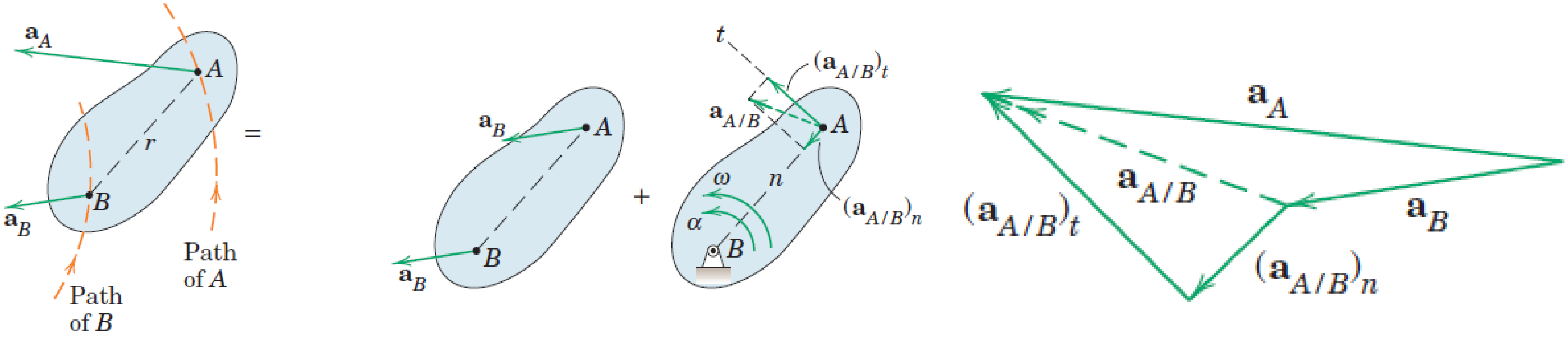
$$\dot{v}_A = \dot{v}_B + \dot{v}_{A/B}$$

$$a_A = a_B + a_{A/B}$$

Relative Acceleration due to Rotation

- If points A and B are located on the same rigid body and in the plane of motion, the distance r between them remains constant so that the observer moving with B perceives A to have circular motion about B
- Because the relative motion is circular, it follows that the relative-acceleration term will have both a normal component directed from A toward B due to the change of direction of $v_{A/B}$ and a tangential component perpendicular to AB due to the change in magnitude of $v_{A/B}$.

$$a_A = a_B + \left(a_{A/B}\right)_n + \left(a_{A/B}\right)_t$$



Relative Acceleration due to Rotation

$$a_A = a_B + \left(a_{A/B}\right)_n + \left(a_{A/B}\right)_t$$

$$\left(a_{A/B}\right)_n = V_{A/B}^2/r = r\omega^2$$

$$\left(a_{A/B}\right)_t = \dot{v}_{A/B} = r\alpha$$

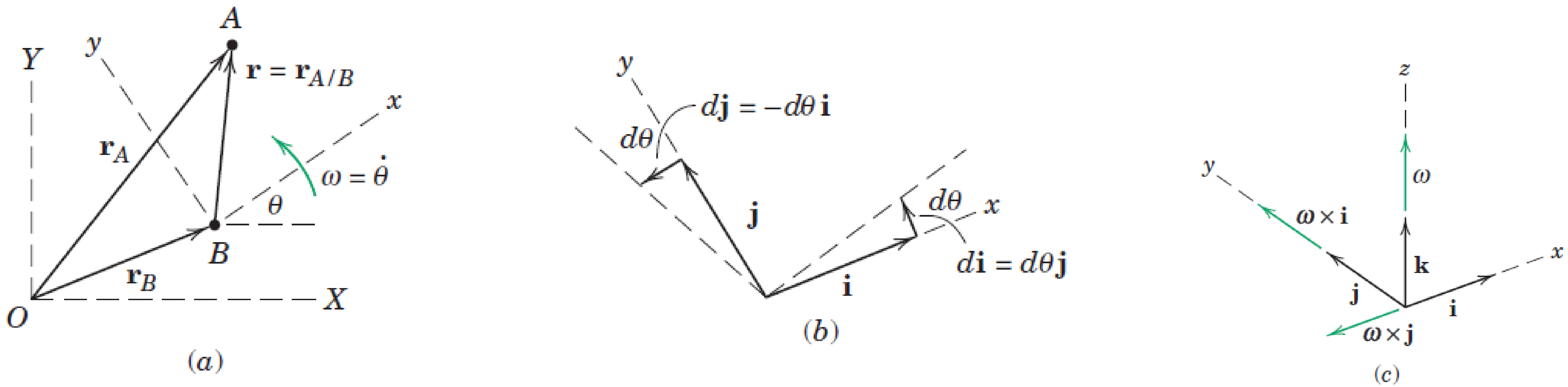
$$\left(\boldsymbol{a}_{A/B}\right)_n = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r})$$

$$\left(\boldsymbol{a}_{A/B}\right)_t = \boldsymbol{\alpha} \times \boldsymbol{r}$$

$\boldsymbol{\omega}$: Absolute angular velocity
$\boldsymbol{\alpha}$: Absolute angular acceleration

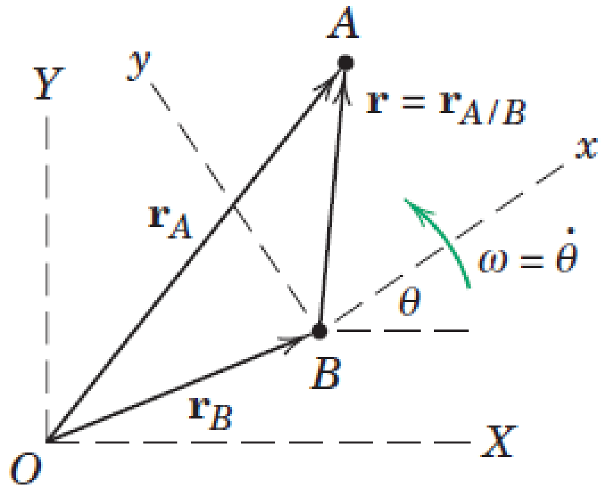
- *Relative* acceleration terms depend on the respective *absolute* angular velocity and *absolute* angular acceleration

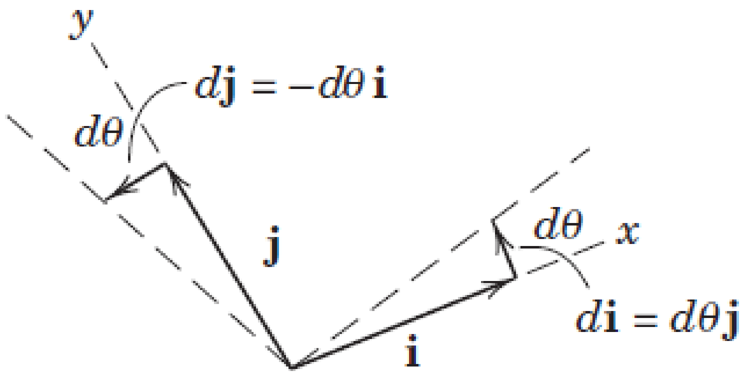
Motion Relative to Rotating Axes

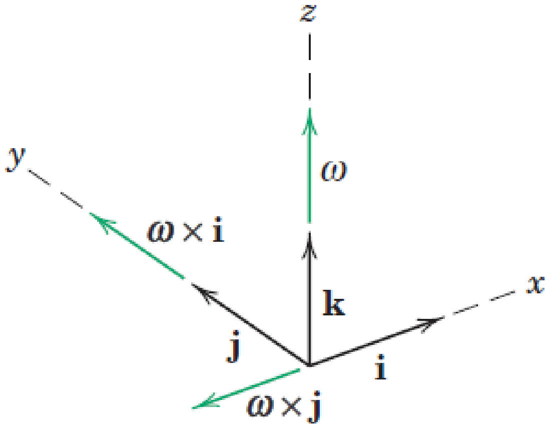


Use of rotating reference axes greatly facilitates the solution of many problems in kinematics where motion is generated within a system or observed from a system which itself is rotating. An example of such a motion is the movement of a fluid particle along the curved vane of a centrifugal pump, where the path relative to the vanes of the impeller becomes an important design consideration

Motion Relative to Rotating Axes

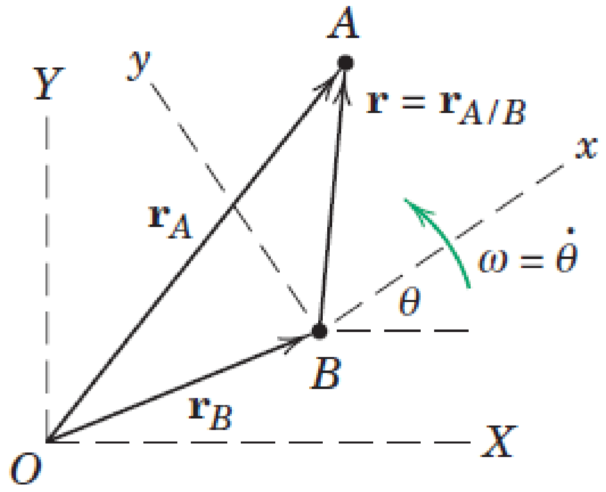


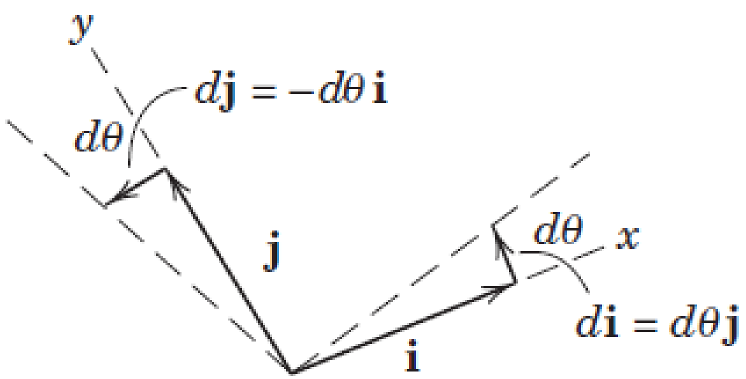


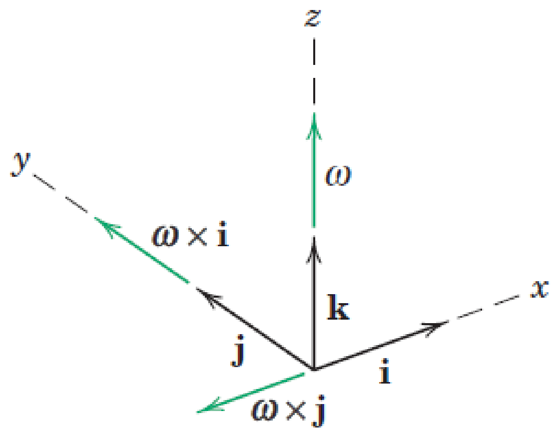


- Description of motion using rotating axes by considering the plane motion of two particles A and B in the fixed X-Y plane.
- A and B to be moving independently of one another
- Motion of A from a moving reference frame x-y which has its origin attached to B and which rotates with an angular velocity $\omega = \dot{\theta}$.
- In vector notation, $\boldsymbol{\omega} = \omega \boldsymbol{k} = \dot{\theta} \boldsymbol{k}$, where the vector is normal to the plane of motion where its positive sense is in the positive z-direction (out from the paper), as established by the right hand rule.

Motion Relative to Rotating Axes





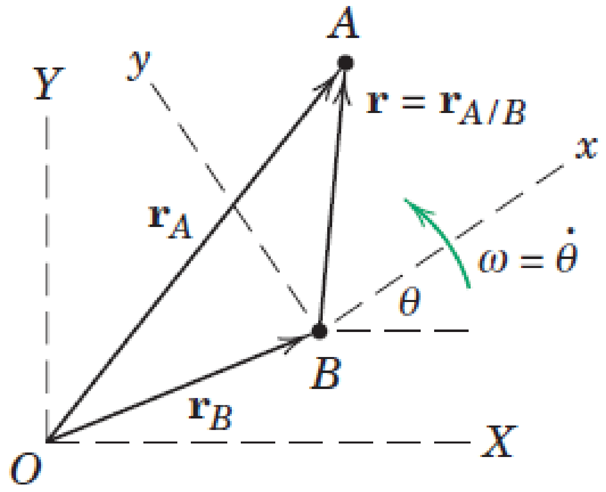


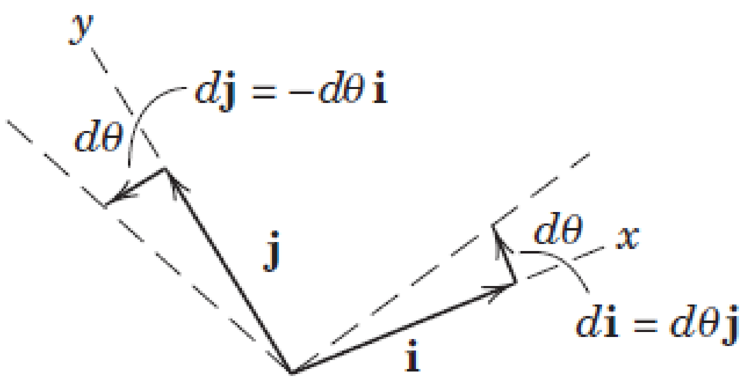
- The absolute position vector of A is given by

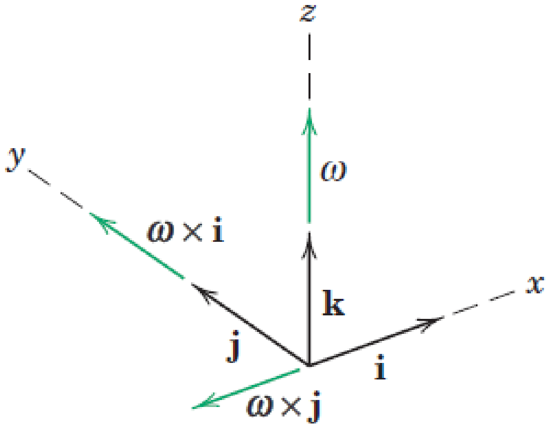
$$\mathbf{r}_A = \mathbf{r}_B + \mathbf{r}$$

where i and j are unit vectors attached to the x-y frame and $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$ stands for $r_{A/B}$, the position vector of A with respect to B.

Time Derivatives of Unit Vectors



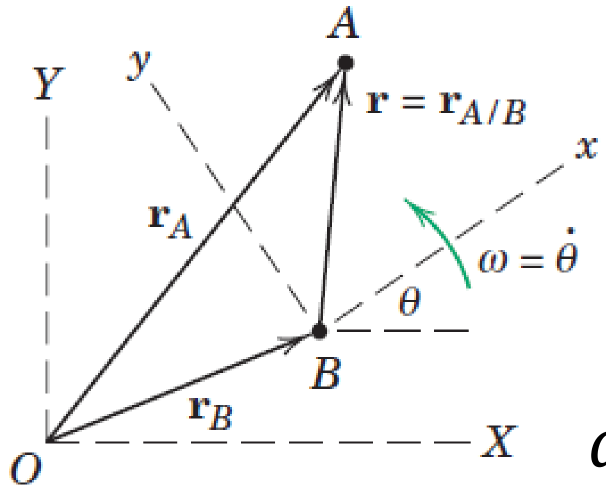


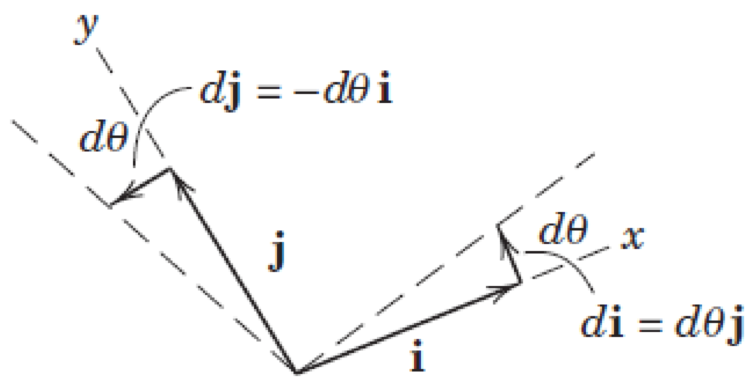


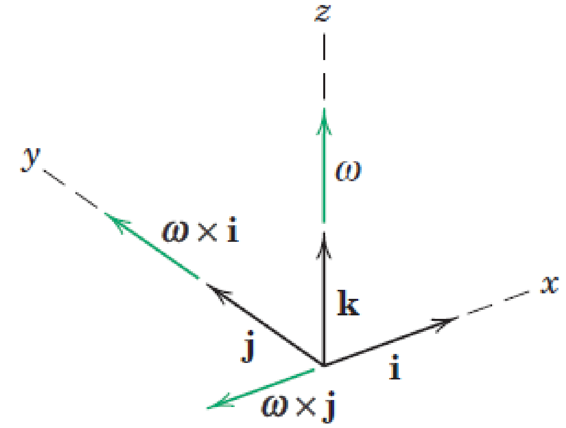
- To obtain the velocity and acceleration equations we must successively differentiate the position-vector equation with respect to time.
- In contrast to the case of translating axes, the unit vectors i and j are now rotating with the x-y axes and, therefore, have time derivatives which must be evaluated.
- The differential change in **i** is d**i**, and it has the direction of **j** and a magnitude equal to the angle d**j** times the length of the vector **i**, which is unity.

$$d\mathbf{i} = d\theta\ \mathbf{j} \qquad\qquad d\mathbf{j} = -\ d\theta\ \mathbf{i}$$

Time Derivatives of Unit Vectors







$$\frac{d\mathbf{i}}{dt} = \frac{d\theta}{dt}\,\mathbf{j}$$

$$\frac{d\mathbf{j}}{dt} = -\frac{d\theta}{dt}\,\mathbf{i}$$

$$\mathbf{i} = \omega\,\mathbf{j}$$

$$\mathbf{j} = -\,\omega\,\mathbf{i}$$

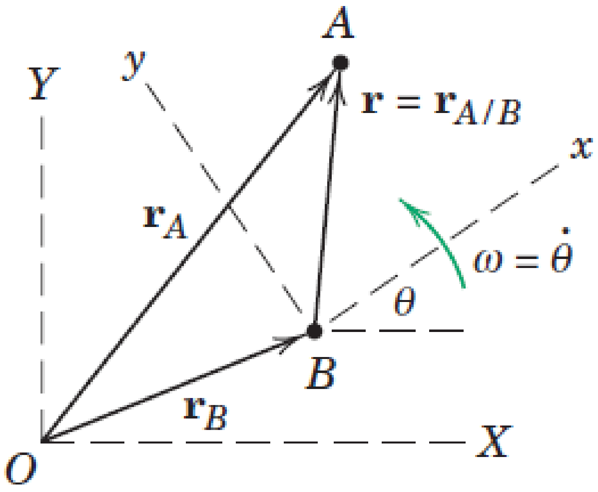
$$\mathbf{i} = \omega\,\mathbf{j} = \boldsymbol{\omega} \times \mathbf{i}$$

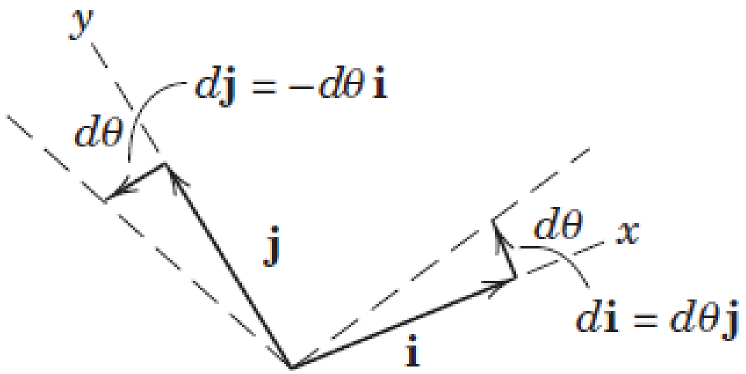
$$\mathbf{j} = -\omega\,\mathbf{i} = \boldsymbol{\omega} \times \mathbf{j}$$

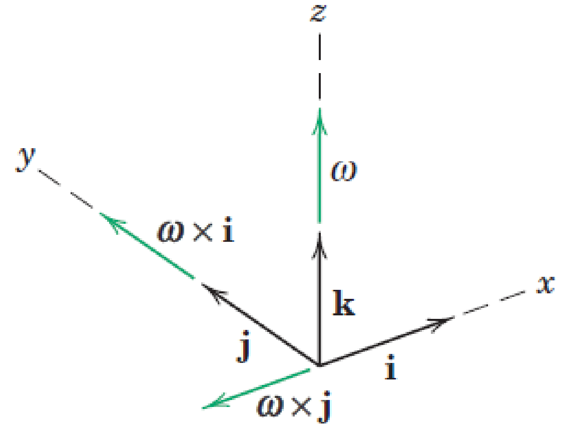
$$\mathbf{i} = \boldsymbol{\omega} \times \mathbf{i}$$

$$\mathbf{j} = \boldsymbol{\omega} \times \mathbf{j}$$

Relative Velocity







$$\mathbf{r}_A = \mathbf{r}_B + \mathrm{x}\mathbf{i} + \mathrm{y}\mathbf{j}$$

$$\dot{\mathbf{i}} = \boldsymbol{\omega} \times \mathbf{i}$$

$$\dot{\mathbf{j}} = \boldsymbol{\omega} \times \mathbf{j}$$

$$\dot{\mathbf{r}}_A = \dot{\mathbf{r}}_B + \dot{\mathrm{x}}\mathbf{i} + \dot{\mathrm{y}}\mathbf{j} + \mathrm{x}\dot{\mathbf{i}} + \mathrm{y}\dot{\mathbf{j}}$$

$$\dot{\mathbf{r}}_A = \dot{\mathbf{r}}_B + \dot{\mathrm{x}}\mathbf{i} + \dot{\mathrm{y}}\mathbf{j} + \mathrm{x}\boldsymbol{\omega} \times \mathbf{i} + \mathrm{y}\boldsymbol{\omega} \times \mathbf{j}$$

$$\dot{\mathbf{r}}_A = \dot{\mathbf{r}}_B + \boldsymbol{\omega} \times \mathrm{x}\mathbf{i} + \boldsymbol{\omega} \times \mathrm{y}\mathbf{j} + \mathbf{v}_{\mathbf{rel}}$$

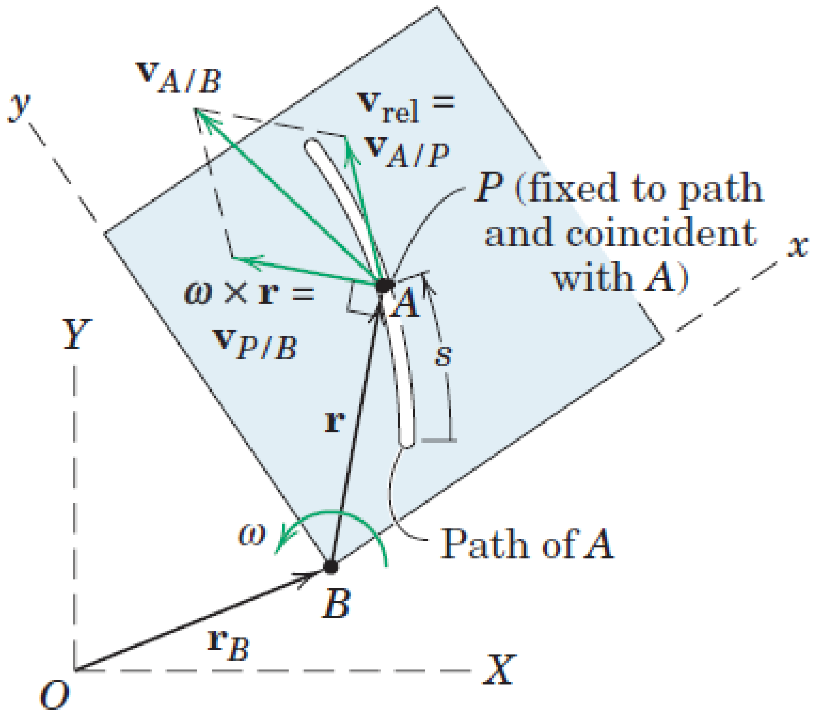
$$\dot{\mathbf{r}}_A = \dot{\mathbf{r}}_B + \boldsymbol{\omega} \times \mathbf{r} + \mathbf{v}_{\mathbf{rel}}$$

$\boldsymbol{\omega} \times \mathbf{r}$ is the difference between the relative velocities as measured from nonrotating and rotating axes

Relative Velocity

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\omega} \times \mathbf{r} + \mathbf{v}_{\mathbf{rel}}$$

- The motion of particle *A* relative to the rotating *x*-*y* plane is shown as taking place in a curved slot in a plate which represents the rotating *x*-*y* reference system.
- The velocity of *A* as measured relative to the plate, $\mathbf{v}_{\mathbf{rel}}$, would be tangent to the path fixed in the *x*-*y* plate and would have a magnitude $\dot{s}$, where s is measured along the path.
- This relative velocity may also be viewed as the velocity $\mathbf{v}_{A/P}$ relative to a point *P* attached to the plate and coincident with *A* at the instant under consideration.
- The term $\boldsymbol{\omega} \times \mathbf{r}$ has a magnitude and a direction normal to $\mathbf{r}$ and is the velocity of point *P* relative to *B* as seen from nonrotating axes attached to *B*.



Relative Velocity

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\omega} \times \mathbf{r} + \mathbf{v}_{\mathbf{rel}}$$

$$\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{P/B} + \mathbf{v}_{A/P}$$

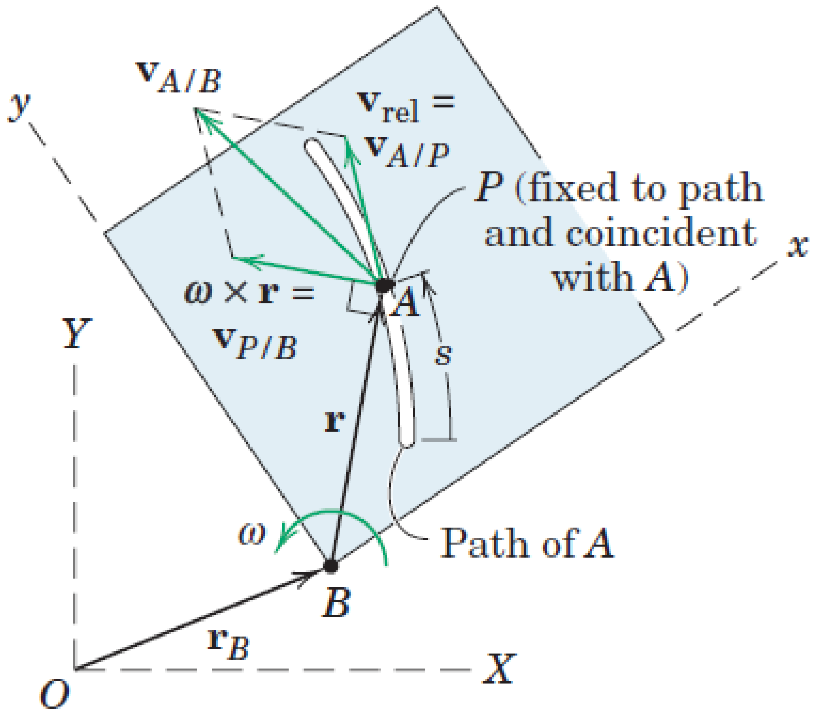
The term $\mathbf{v}_{P/B}$ is measured from a nonrotating position.

Term $\mathbf{v}_{A/P}$ is the same as $\mathbf{v}_{\mathbf{rel}}$ and is the velocity of A as measured in the x-y frame.

$$\mathbf{v}_A = \mathbf{v}_P + \mathbf{v}_{A/P}$$

$\mathbf{v}_P$ is the absolute velocity of P and represents the effect of the moving coordinate system, both translational and rotational

$$\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{A/B}$$



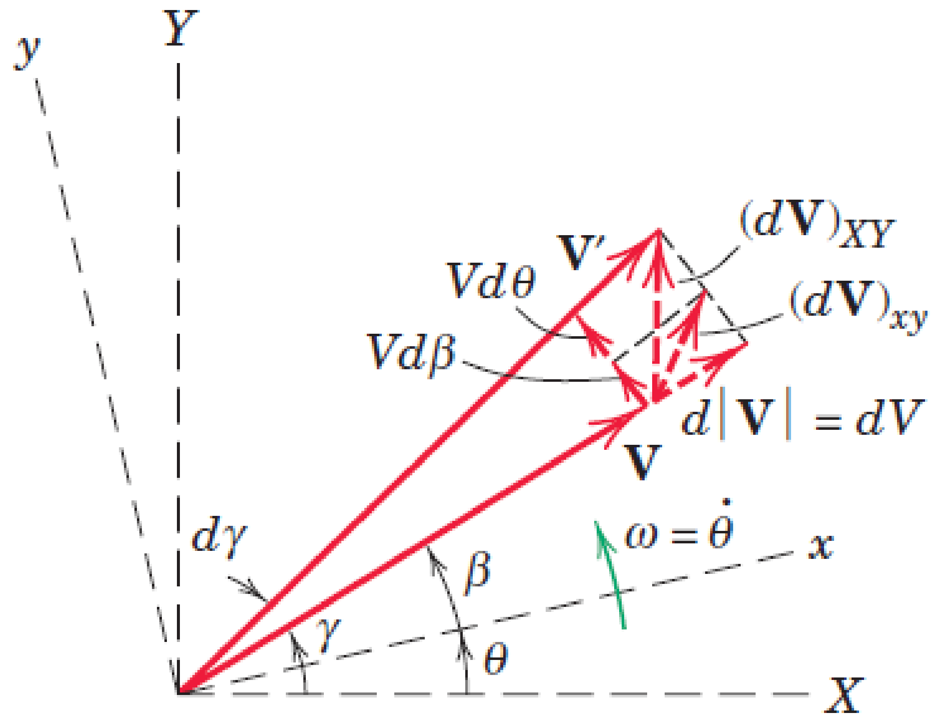
Transformation of a Time Derivative

$$\left(\frac{\mathrm{d}\mathbf{V}}{\mathrm{dt}}\right)_{\mathrm{XY}} = (\dot{\mathrm{V}}_x\mathrm{i} + \dot{\mathrm{V}}_y\mathrm{j}) + (V_x\dot{\mathrm{i}} + V_y\dot{\mathrm{j}})$$

- First two terms in the expression represent that part of the total derivative of **V** which is measured relative to the *x*-*y* reference system,
- Second two terms represent that part of the derivative due to the rotation of the reference system.

$$\left(\frac{\mathrm{d}\mathbf{V}}{\mathrm{dt}}\right)_{\mathrm{XY}} = \left(\frac{\mathrm{d}\mathbf{V}}{\mathrm{dt}}\right)_{\mathrm{xy}} + \boldsymbol{\omega} \times \mathbf{V}$$

$\boldsymbol{\omega} \times \mathbf{V}$ represents the difference between the time derivative of the vector as measured in a fixed reference system and its time derivative as measured in the rotating reference system



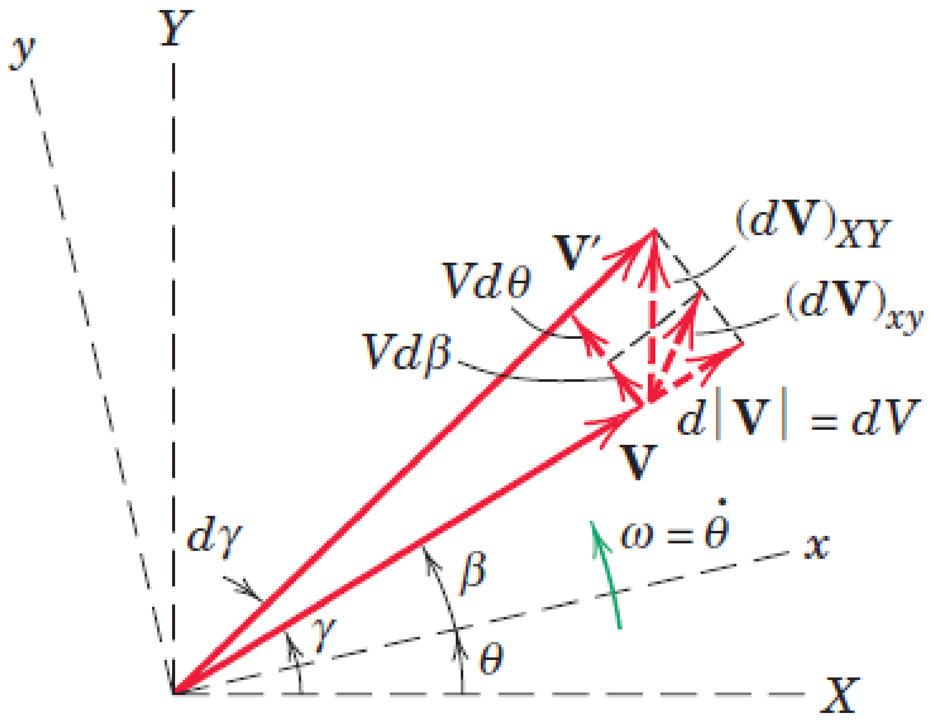
Transformation of a Time Derivative

- Figure shows the vector $\mathbf{V}$ at time t as observed both in the fixed axes X-Y and in the rotating axes x-y.

- During time dt, the vector swings to position $\mathbf{V}'$, and the observer in x-y measures the two components (a) dV due to its change in magnitude and (b) $Vd\beta$ due to its rotation $d\beta$ relative to x-y.

- To the rotating observer, then, the derivative $\frac{dV}{dt}_{\text{xy}}$ which the observer measures has the components $\frac{dV}{dt}$ and $V\frac{d\beta}{dt} = V\dot{\beta}$.

- The remaining part of the total time derivative not measured by the rotating observer has the magnitude $V\frac{d\theta}{dt}$ and, expressed as a vector, is $\boldsymbol{\omega} \times \mathbf{V}$.



$$\dot{\mathbf{V}}_{\text{XY}} = \dot{\mathbf{V}}_{\text{xy}} + \boldsymbol{\omega} \times \mathbf{V}$$

Relative Acceleration

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\omega} \times \mathbf{r} + \mathbf{v}_{\mathbf{rel}}$$

$$\dot{\mathbf{v}}_A = \dot{\mathbf{v}}_B + \dot{\boldsymbol{\omega}} \times \mathbf{r} + \boldsymbol{\omega} \times \dot{\mathbf{r}} + \dot{\mathbf{v}}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \dot{\boldsymbol{r}} + \dot{\mathbf{v}}_{\mathbf{rel}}$$

$$\dot{\boldsymbol{r}} = \frac{d(x\mathrm{i} + y\mathrm{j})}{dt} = x\dot{\mathbf{i}} + y\dot{\mathbf{j}} + \dot{x}\mathbf{i} + \dot{y}\mathbf{j} = \boldsymbol{\omega} \times \boldsymbol{r} + \mathbf{v}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r} + \mathbf{v}_{\mathbf{rel}}) + \dot{\mathbf{v}}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + \boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \dot{\mathbf{v}}_{\mathbf{rel}}$$

Relative Acceleration

$$\mathbf{v}_{\mathbf{rel}} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j}$$

$$\dot{\mathbf{v}}_{\mathbf{rel}} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j} + \boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} = \boldsymbol{a}_{\mathbf{rel}} + \boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + \boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \dot{\mathbf{v}}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + \boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}} + \boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}}$$

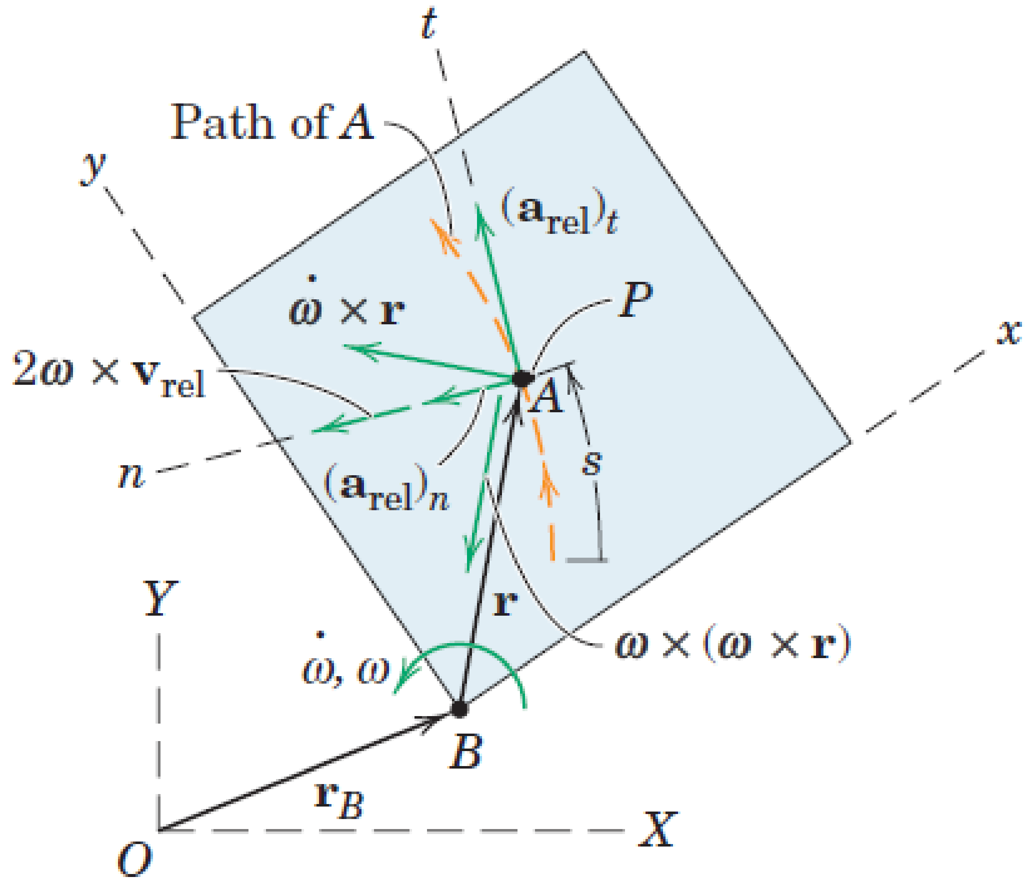
$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

Relative Acceleration

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$$
$$+2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

Above equation is the general vector expression for the absolute acceleration of a particle *A* in terms of its acceleration $\boldsymbol{a}_{\mathbf{rel}}$ measured relative to a moving coordinate system which rotates with an angular velocity $\boldsymbol{\omega}$ and an angular acceleration $\dot{\boldsymbol{\omega}}$.

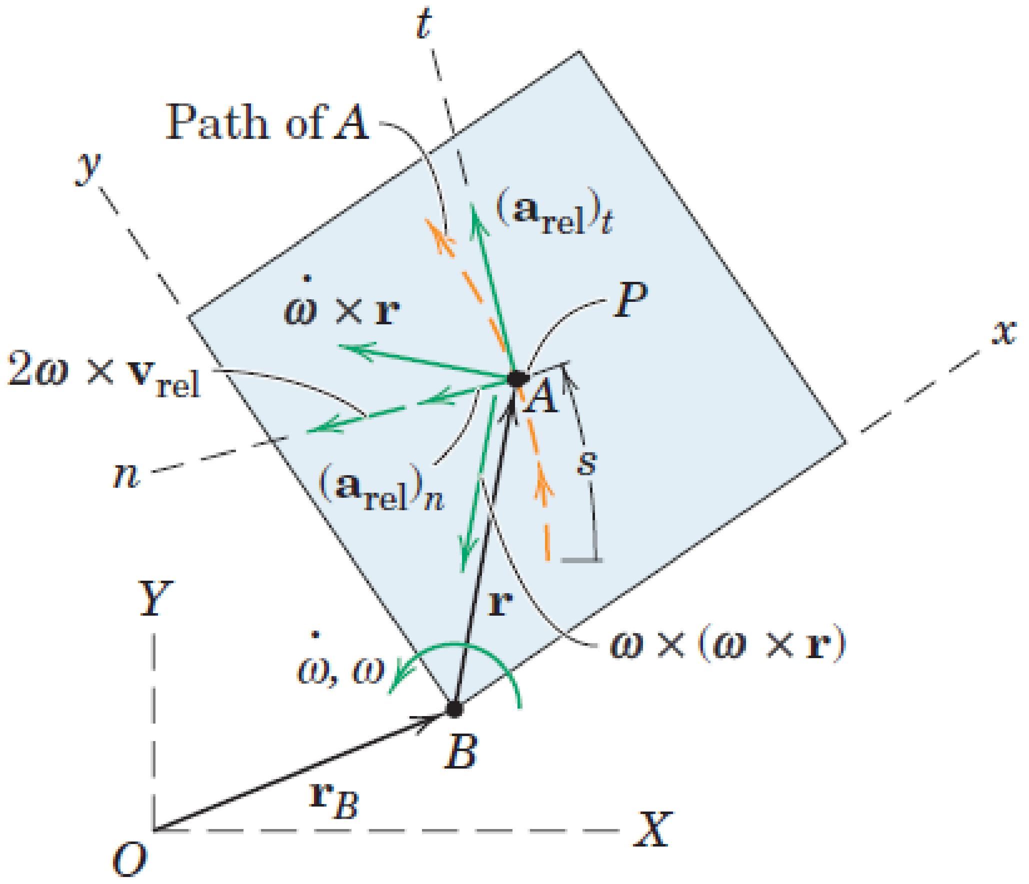
- $\dot{\boldsymbol{\omega}} \times \boldsymbol{r}$ Tangential component of the acceleration $\boldsymbol{a}_P$ of the coincident point *P* in its circular motion with respect to *B*.
- $\boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$ normal component of the acceleration $\boldsymbol{a}_P$ of the coincident point *P* in its circular motion with respect to *B*.



Relative Acceleration

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$$
$$+2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

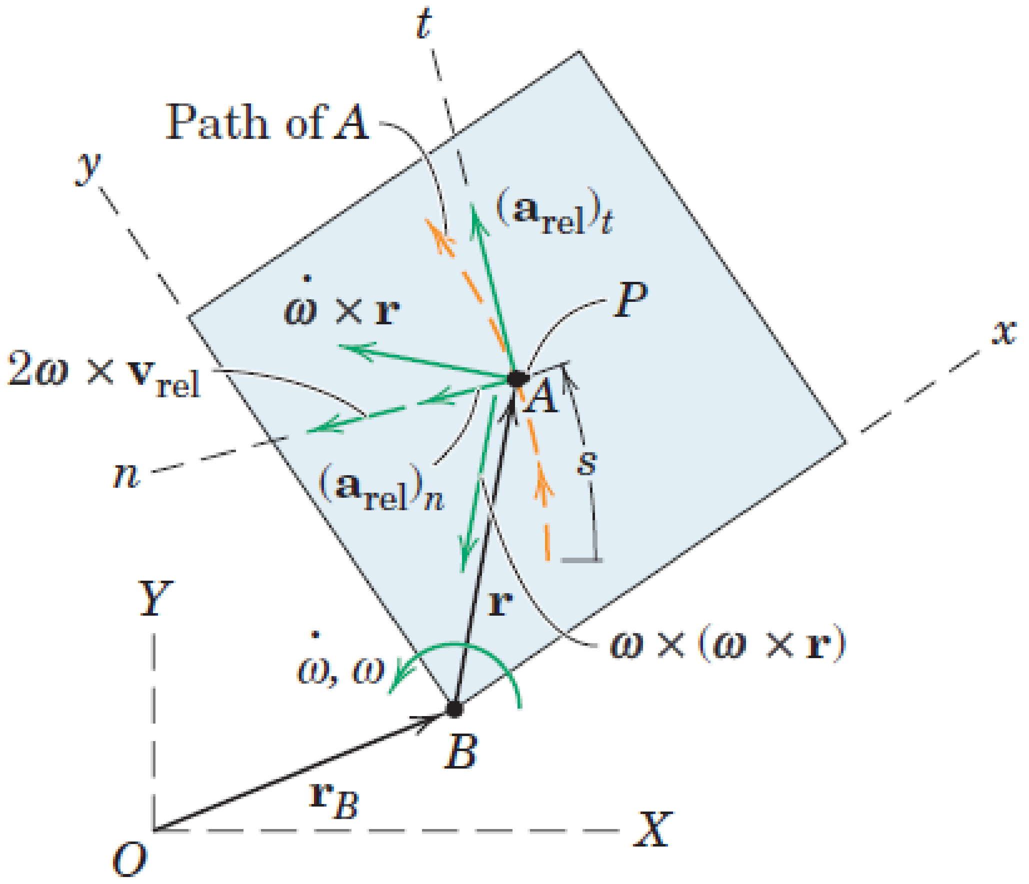
- $\dot{\boldsymbol{\omega}} \times \boldsymbol{r}$ Tangential component of the acceleration $\boldsymbol{a}_P$ of the coincident point *P* in its circular motion with respect to *B*.
- $\boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$ normal component of the acceleration $\boldsymbol{a}_P$ of the coincident point *P* in its circular motion with respect to *B*.
- This motion would be observed from a set of nonrotating axes moving with B.
- The magnitude of $\dot{\boldsymbol{\omega}} \times \boldsymbol{r}$ is $r\ddot{\theta}$ and its direction is tangent to the circle. The magnitude of $\boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$ is $\boldsymbol{r}\boldsymbol{\omega}^2$ and its direction is from P to B along the normal to the circle.



Relative Acceleration

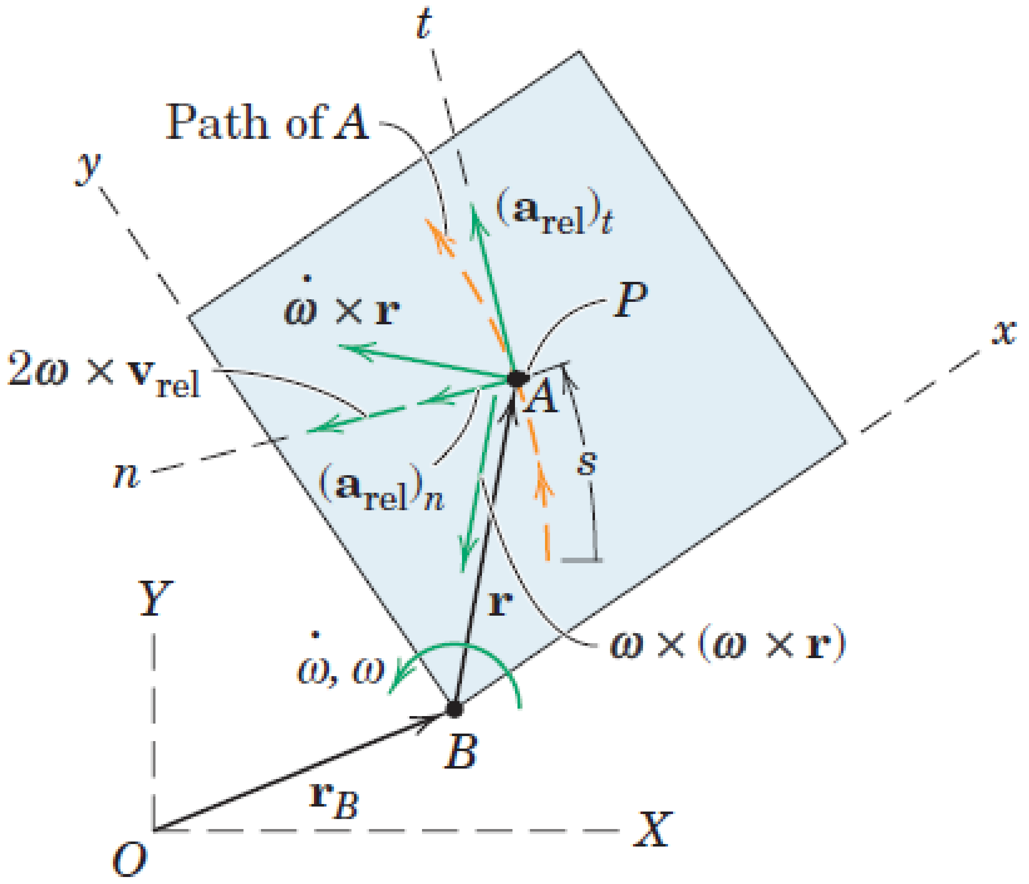
$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$$
$$+ 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

- The acceleration of *A* relative to the plate along the path, $\boldsymbol{a}_{\mathbf{rel}}$, may be expressed in rectangular, normal and tangential, or polar coordinates in the rotating system. Frequently, *n*- and *t*-components are used.
- The tangential component has the magnitude $\boldsymbol{a}_{\mathbf{rel}_t} = \ddot{s}$, where s is the distance measured along the path to *A*.
- The normal component has the magnitude $\boldsymbol{a}_{\mathbf{rel}_n} = \frac{v_{rel}^2}{\rho}$, where ρ is the radius of curvature of the path as measured in *x*-*y*. The sense of this vector is always toward the center of curvature.



Coriolis Acceleration

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$



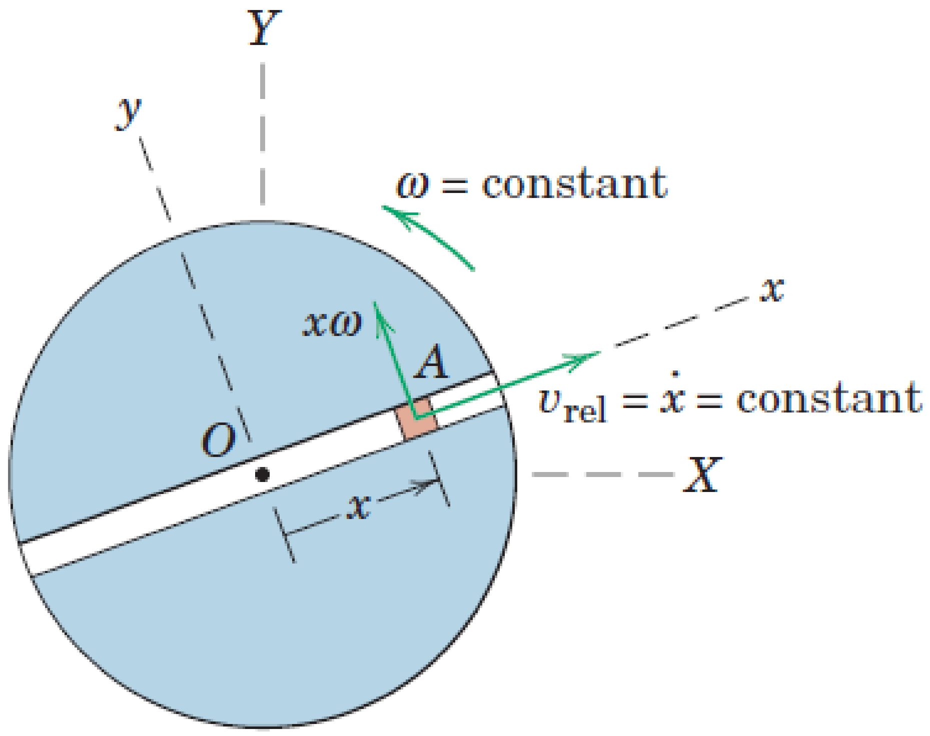
- The term $2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}}$ is called the ***Coriolis* acceleration**.
- Named after the French military engineer G. Coriolis (1792–1843), who was the first to call attention to this term.
- It represents the difference between the acceleration of *A* relative to *P* as measured from nonrotating axes and from rotating axes.
- The direction is always normal to the vector $\mathbf{v}_{\mathbf{rel}}$, and the sense is established by the right-hand rule for the cross product.

Coriolis Acceleration

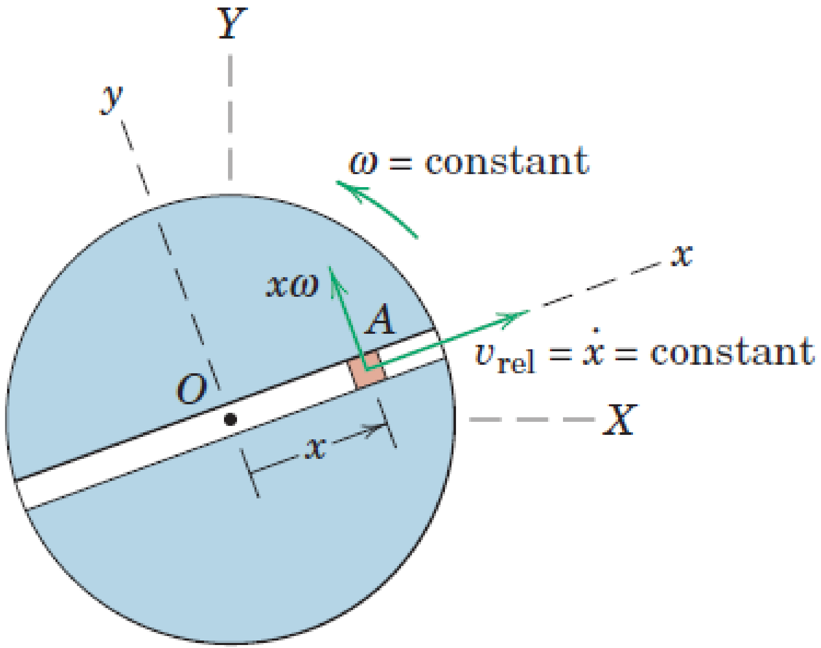
$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$$
$$+2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

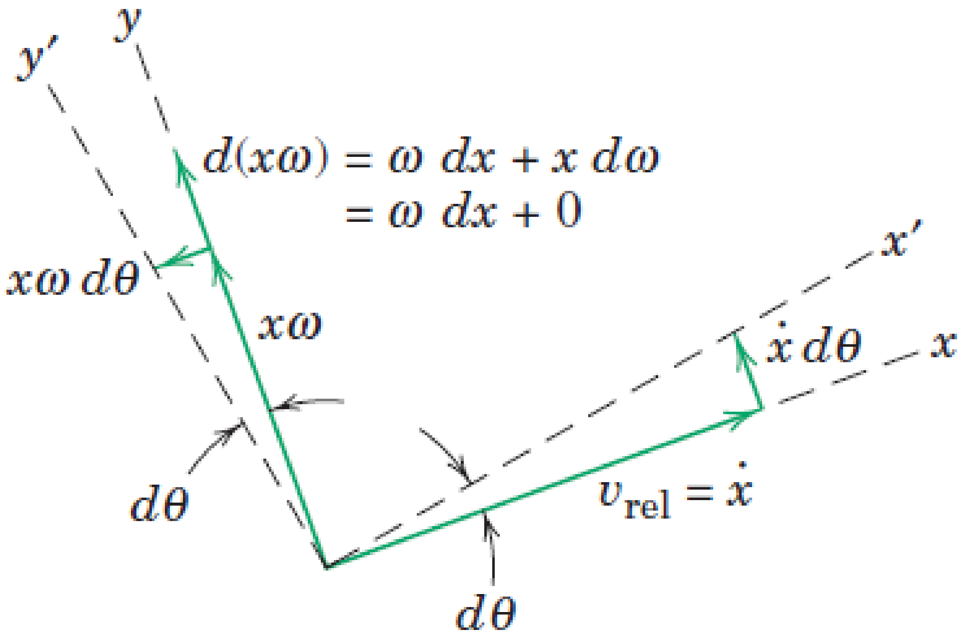
- The Coriolis acceleration $2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}}$ is difficult to visualize because it is composed of two separate physical effects. To help with this visualization, we will consider the simplest possible motion in which this term appears.

- We have a rotating disk with a radial slot in which a small particle *A* is confined to slide. Let the disk turn with a constant angular velocity $\boldsymbol{\omega}$ and let the particle move along the slot with a constant speed $v_{rel} = \dot{x}$ relative to the slot. The velocity of *A* has the two components (*a*) $\dot{x}$ due to motion along the slot and (*b*) $x\omega$ due to the rotation of the slot.



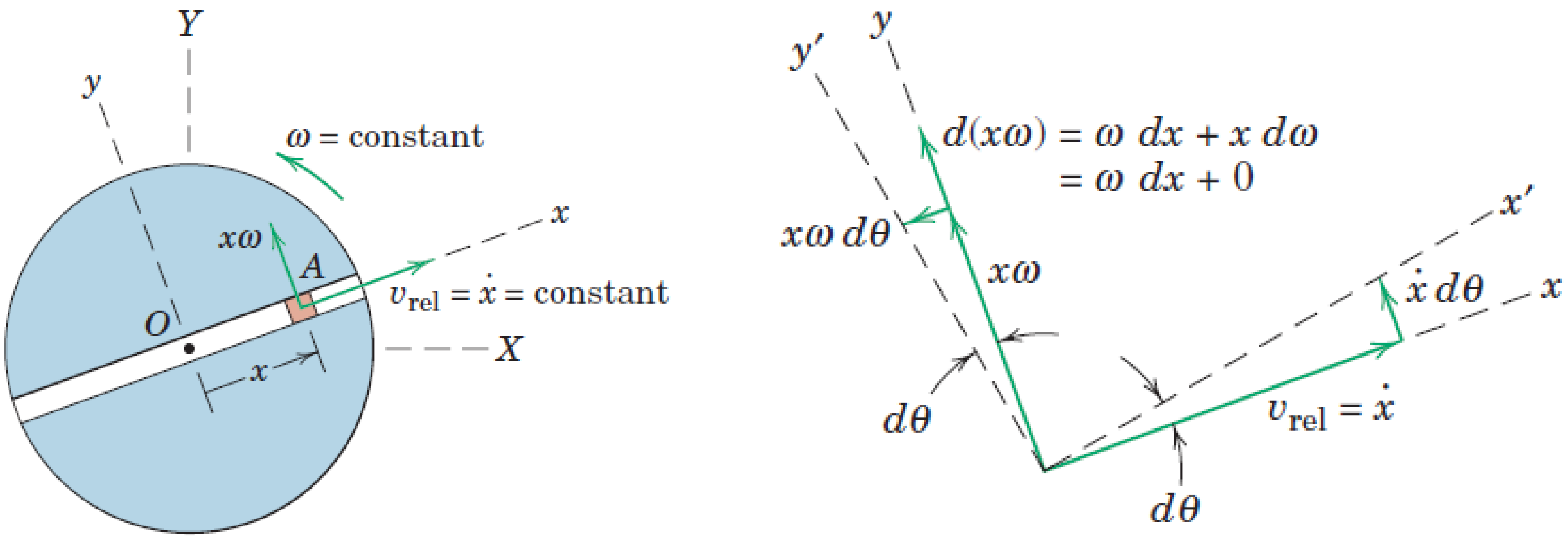
Coriolis Acceleration





- The changes in these two velocity components due to the rotation of the disk are shown in 2nd figure for the interval dt, during which the *x*-*y* axes rotate with the disk through the angle $d\theta$ to *x'*-*y'*.
- The velocity increment due to the change in direction of v_{rel} is $\dot{x}d\theta$ and that due to the change in magnitude of $x\omega$ is ωdx, both being in the y-direction normal to the slot.

Coriolis Acceleration

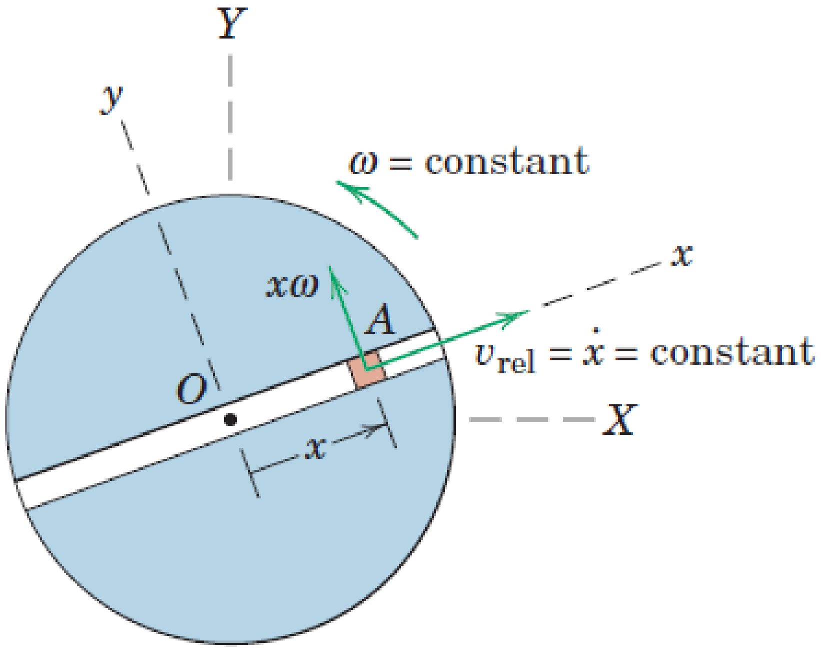


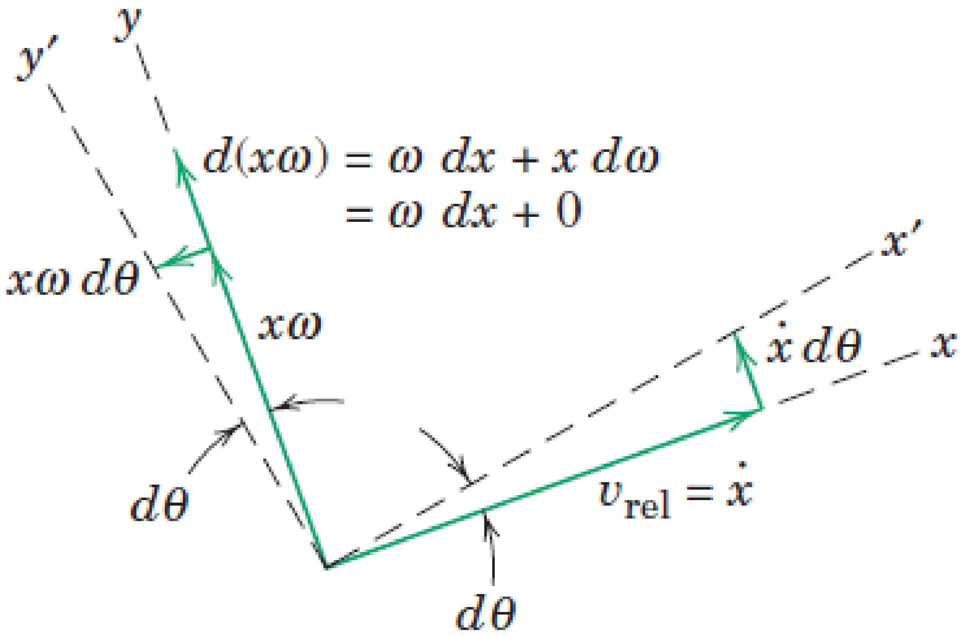
- The velocity increment due to the change in direction of v_{rel} is $\dot{x}d\theta$ and that due to the change in magnitude of $x\omega$ is ωdx, both being in the y-direction normal to the slot.

$$\frac{\dot{x}d\theta + \omega dx}{dt} = \dot{x}\dot{\theta} + \omega\dot{x} = \omega\dot{x} + \omega\dot{x} = 2\omega\dot{x}$$

- For this case, $2\omega\dot{x}$ is the magnitude of the Coriolis acceleration $\mathbf{2\boldsymbol{\omega}} \times \mathbf{v_{rel}}$

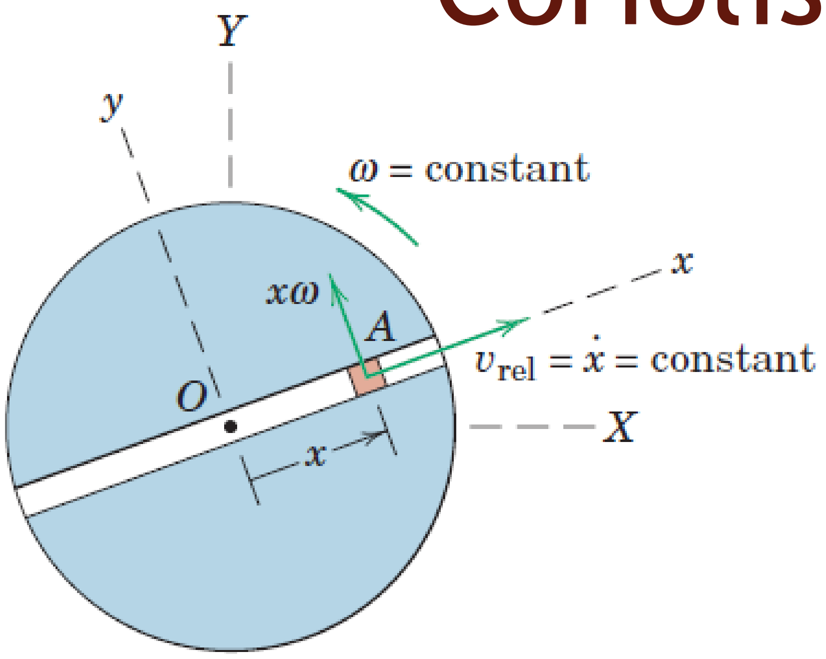
Coriolis Acceleration

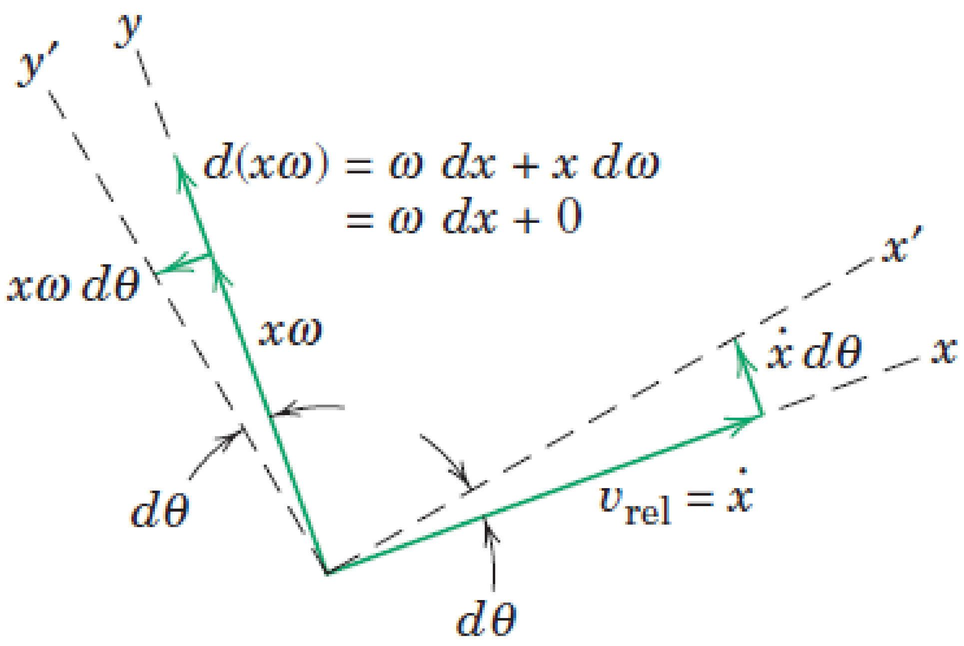




- Dividing the remaining velocity increment $x\omega d\theta$ due to the change in direction of $x\omega$ by dt gives $x\omega\dot{\theta}$ or $x\omega^2$, which is the acceleration of a point *P* fixed to the slot and momentarily coincident with the particle *A*.

Coriolis Acceleration





$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

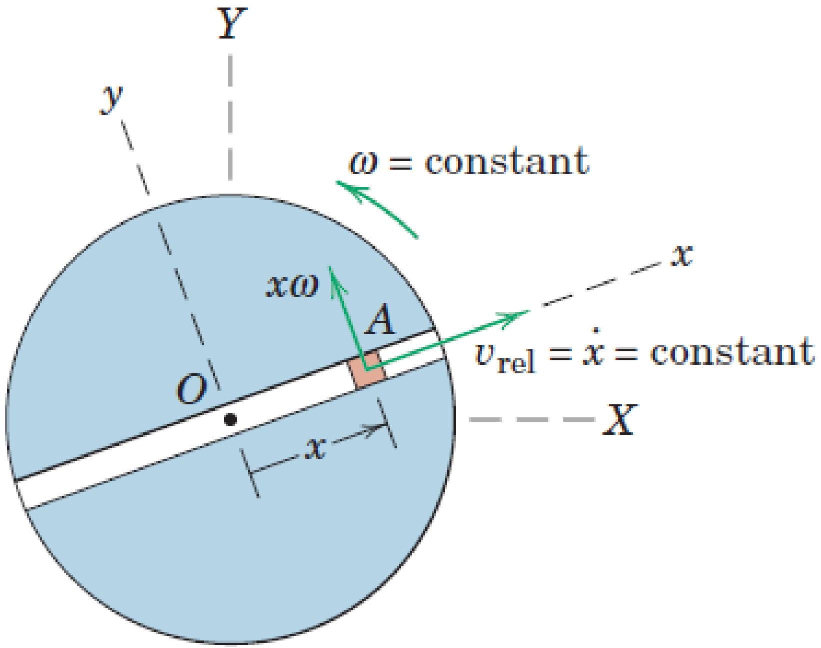
- With the origin *B* in that equation taken at the fixed center *O*, $\boldsymbol{a}_B = 0$.
- With constant angular velocity, $\dot{\boldsymbol{\omega}} \times \boldsymbol{r} = \mathbf{0}$
- With $\mathbf{v}_{\mathbf{rel}}$ constant in magnitude and no curvature to the slot, $\boldsymbol{a}_{\mathbf{rel}} = 0$

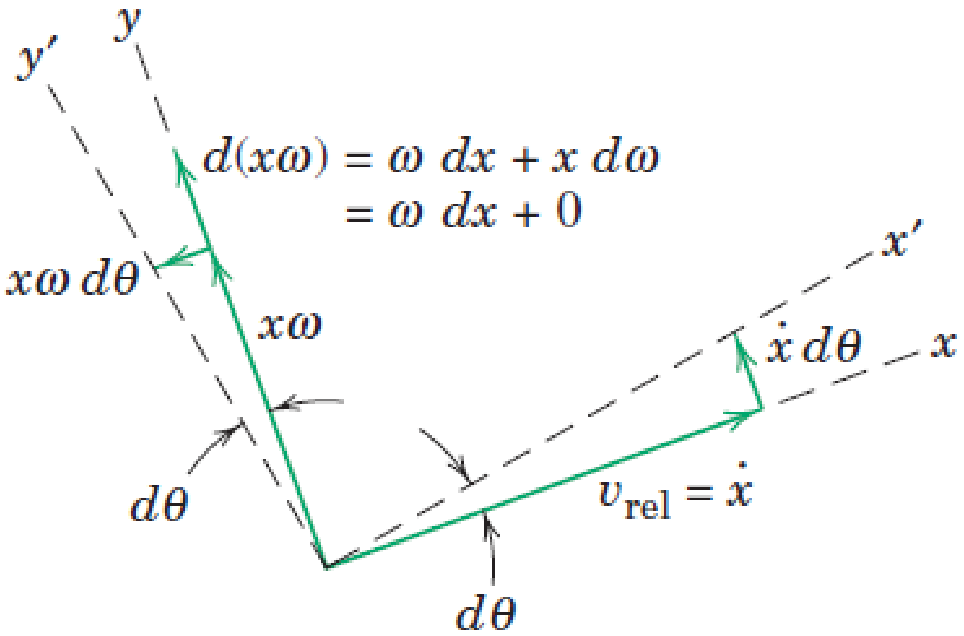
$$\boldsymbol{a}_A = \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}}$$

- Replacing $\boldsymbol{r}$ by $x\mathbf{i}$, $\boldsymbol{\omega}$ by $\omega\mathbf{k}$, and $\mathbf{v}_{\mathbf{rel}}$ by $\dot{x}\mathbf{i}$ gives

$$a_A = -x\omega^2 \mathrm{i} + 2\dot{x}\omega j$$

Coriolis Acceleration





- Replacing $\boldsymbol{r}$ by $x\mathbf{i}$, $\boldsymbol{\omega}$ by $\omega\mathbf{k}$, and $\mathbf{v}_{\mathbf{rel}}$ by $\dot{x}\mathbf{i}$ gives

$$a_A = -x\omega^2 \mathrm{i} + 2\dot{x}\omega j$$

- If the slot in the disk had been curved, we would have had a normal component of acceleration relative to the slot so that $\boldsymbol{a}_{\mathbf{rel}}$ would not be zero.

Rotating vs Non-Rotating Systems

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \qquad \boldsymbol{a}_{\boldsymbol{P}/B} \qquad + \qquad \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_P + \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \boldsymbol{a}_{A/B}$$

- From the third equation where $\boldsymbol{a}_B$, $\boldsymbol{a}_{\boldsymbol{P}/B}$ has been combined to give $\boldsymbol{a}_{\boldsymbol{P}}$, it is seen that the relative-acceleration term $\boldsymbol{a}_{A/P}$, unlike the corresponding relative-velocity term, is not equal to the relative acceleration $\boldsymbol{a}_{\mathbf{rel}}$ measured from the rotating *x*-*y* frame of reference.

- The Coriolis term $(2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}})$ is, therefore, the difference between the acceleration $\boldsymbol{a}_{A/P}$ of *A* relative to *P* as measured in a nonrotating system and the acceleration $\boldsymbol{a}_{\mathbf{rel}}$ of *A* relative to *P* as measured in a rotating system.

Rotating vs Non-Rotating Systems

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \qquad \boldsymbol{a}_{\boldsymbol{P}/B} \qquad + \qquad \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_P + \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \boldsymbol{a}_{A/B}$$

- From the fourth equation, it is seen that the acceleration $\boldsymbol{a}_{A/B}$ of A with respect to B as measured in a nonrotating system is a combination of the last four terms in the first equation for the rotating system.

Rotating vs Non-Rotating Systems

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \qquad \boldsymbol{a}_{\boldsymbol{P}/B} \qquad + \qquad \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_P + \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \boldsymbol{a}_{A/B}$$

- Writing the acceleration of *A* in terms of the acceleration of the coincident point *P*

$$\boldsymbol{a}_A = \boldsymbol{a}_P + \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_P + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

- Because the acceleration of *P* is $\boldsymbol{a}_P = \boldsymbol{a}_B + \boldsymbol{a}_{\boldsymbol{P}/B}$
- When the equation is written in this form, point *P* may not be picked at random because it is the one point attached to the rotating reference frame coincident with *A* at the instant of analysis.

Rotating vs Non-Rotating Systems

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \qquad \boldsymbol{a}_{\boldsymbol{P}/B} \qquad + \qquad \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_P + \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \boldsymbol{a}_{A/B}$$

$\mathbf{v}_B$: absolute velocity of the origin *B* of the rotating axes
$\boldsymbol{a}_B$: absolute acceleration of the origin *B* of the rotating axes
$\boldsymbol{r}$: position vector of the coincident point *P* measured from *B*
$\boldsymbol{\omega}$: angular velocity of the rotating axes
$\dot{\boldsymbol{\omega}}$: angular acceleration of the rotating axes
$\mathbf{v}_{\mathbf{rel}}$: velocity of *A* measured relative to the rotating axes
$\boldsymbol{a}_{\mathbf{rel}}$: acceleration of *A* measured relative to the rotating axes

THANK YOU